DESIGN AND IMPLEMENTATION OF A WEB-BASED CLEARANCE SYSTEM FOR GRADUATING STUDENTS IN BABCOCK UNIVERSITY

BY

AIYEDUN TREASURE TENIOLA – 21/0743

IGWE NOBLE CHISOM – 21/0530

ODITA CHUKWUDUMEBI TOBECHUKWU – 21/1407

A PROJECT SUBMITTED IN PARTIAL FULFILLMENT TO THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE B.SC(HONS) DEGREE IN SOFTWARE ENGINEERING

TO THE DEPARTMENT OF SOFTWARE ENGINEERING, SCHOOL OF COMPUTING, BABCOCK UNIVERSITY, ILISHAN REMO, OGUN STATE, NIGERIA.

APRIL, 2025.

DECLARATION

We declare that this project "Design and Implementation of a Web-Based Clearance System for Graduating Students" was carried out by the following people:

AIYEDUN TREASURE TENIOLA	_____
21/0743	**SIGNATURE/DATE**
IGWE CHISOM NOBLE	_____
21/0530	**SIGNATURE/DATE**
ODITA CHUKWUDUMEBI	_____
21/1407	**SIGNATURE/DATE**

CERTIFICATION

I certify that this project was carried out by the following students under my supervision in the department of Software Engineering, Babcock University, Ilishan-Remo, Ogun State, Nigeria.

AIYEDUN TREASURE TENIOLA	**21/0743**
ODITA CHUKWUDUMEBI TOBECHUKWU	**21/1407**
IGWE NOBLE CHISOM	**21/0530**

DR. EWEOYA _____

(Project Supervisor) **DATE**

EXTERNAL EXAMINER _____

DATE

DR. SOLA MAITAIMI _____

(Head of Department) **DATE**

DEDICATION

First and foremost, in humble reverence to the Almighty God, we thank him for blessing us with wisdom, knowledge, good health, and insight necessary for completing this work. This project is also dedicated to our parents, family, friends, and loved ones whose unwavering support and engagement have been a constant source of strength. To our mentors and educators, whose guidance and wisdom have shaped our academic journey, we dedicate this project as a testament to your dedication.

ACKNOWLEDGEMENT

We would like to express our deep gratitude to the Almighty God, whose guidance and blessings have been ever-present throughout this research journey. We are grateful to our families and friends for their love, patience, and understanding during this time. Their unwavering support has been a source of strength and inspiration. We would like to especially thank our loving and wonderful parents in the same of Mr. and Mrs. Aiyedun, Mr. and Mrs. Igwe, and Mr. and Mrs. Odita for their amazing love. We extend our heartfelt thanks to our supervisor, Dr. Eweoya, for his unwavering support, guidance, and encouragement. His expertise and insights have been invaluable in shaping this project. Lastly, to everyone who has played a part in this project, whether big or small, we offer our heartfelt thanks. Your support and encouragement have been truly appreciated.

ABSTRACT

The clearance process for graduating students has traditionally relied on manual, paper-based procedures, which are often time-consuming, prone to errors, and lack real-time communication between departments. These inefficiencies result in delays, misplaced documents, and bottlenecks in the issuance of certificates and transcripts. This project focuses on the development of a web-based clearance system to address these challenges by streamlining the clearance process. The system centralizes the clearance activities across various departments, allowing students to monitor their progress in real-time and complete clearance tasks efficiently. Developed using ExpressJS for the backend and ReactJS for the frontend, the system integrates key functionalities such as payment verification, document submission, and automated notifications. Unified Modeling Language (UML) diagrams were used to model the system's functional requirements. The results demonstrate that the proposed system significantly reduces clearance processing time, minimizes human errors, and improves administrative efficiency. This approach ensures cost savings, eliminates unnecessary delays, and enhances the overall graduation experience for students while maintaining the integrity and reliability of the clearance process.

TABLE OF CONTENT

CHAPTER ONE 1

1.0 Background of The Study 1

1.1 Statement of The Problem 3

1.2 Aim and Specific Objectives 4

1.3 Methodology 4

1.4 Scope of The Study 5

1.5 Significance of the Study 6

1.6 Definition of Terms 6

1.7 Organization of The Project 7

CHAPTER TWO 8

2.1 Historical Background of the Research 8

2.2 Conceptual Overview 12

2.2.1 Definition of Web – Based Clearance and Web – Based Clearance System 12

2.2.2 Key Components of a Web-Based Clearance Systems for Graduating Students 13

2.2.3 Data Types in Web-Based Online Clearance Systems for Graduating Students 15

2.2.4 General Functions and Operations of a Web-Based Clearance Systems for Graduating Students 16

2.2.5 Applications of a Web-Based Clearance Systems for Graduating Students in Real Life 16

2.2.6 Challenges and Considerations **17**

2.3 Overview of Existing Systems **17**

2.4 Review of Related Work **19**

2.4.1 Summary of Reviewed Literature **22**

2.5 Strength and Weaknesses of the Existing Systems **22**

CHAPTER THREE **24**

3.0 Introduction **24**

3.1 Research Design **24**

3.2 Software Development Process Model **25**

3.3 Requirements Elicitation **27**

3.3.1 User Requirements **27**

3.3.2 Functional Requirements **28**

3.3.3 Non-Functional Requirements **28**

3.3.4 System Requirements **30**

3.3.4.1 Hardware Requirements 30

3.3.4.2 Software Requirements 30

3.4 System Architecture **30**

3.5 System Design **31**

3.6 Development Tools **37**

CHAPTER FOUR **38**

4.0 Introduction **38**

4.1 System Architecture **38**

4.1.1 System Components **38**

4.2 System Implementation **39**

4.2.1 Signup and Login Page **39**

4.2.2 Student Dashboard **40**

4.2.3 Staff/Office Dashboard **42**

4.2.4 Admin Dashboard **44**

4.3 Implementation Tools and Technologies **46**

4.4 User Roles and Permissions **47**

4.5 System Workflow **47**

CHAPTER FIVE **48**

5.0 Summary of Findings **48**

5.1 Limitations **49**

5.2 Conclusion **49**

5.3 Recommendations **50**

REFERENCES **51**

LIST OF FIGURES

Figure 3.1 – Architectural Diagram …… 31

Figure 3.2 – Flowchart Diagram …… 33

Figure 3.3 – Use Case Diagram …….... 34

Figure 3.4 – Sequence Diagram …… 35

Figure 3.5 – Entity Relationship Diagram(ERD) ……..…….. 36

Figure 4.1 – Signup Page …… 39

Figure 4.2 – Login Page ……..... 40

Figure 4.3 – Student Dashboard ……..... 41

Figure 4.4 – Staff Dashboard ……. 42

Figure 4.5 – Admin Dashboard …….. 45

TURNITIN ORIGINALITY REPORT

Turnitin Originality Report

Processed on: 16-Mar-2025 12:39 PM CDT
ID: 2616069875
Word Count: 8768
Submitted: 1

final year project By Treasure Aiyedun

Similarity Index	Similarity by Source	
11%	Internet Sources:	7%
	Publications:	2%
	Student Papers:	7%

1% match (student papers from 10-Apr-2023)
Submitted to CVC Nigeria Consortium on 2023-04-10

1% match (Internet from 28-Feb-2025)
https://sparklyn.com.ng/online-clearance-system-for-graduating-students-16950-d

< 1% match (student papers from 03-Apr-2017)
Submitted to CVC Nigeria Consortium on 2017-04-03

< 1% match (student papers from 11-Apr-2023)
Submitted to CVC Nigeria Consortium on 2023-04-11

< 1% match (publications)
Prabh Deep Singh, Mohit Angurala. "Integration of Cloud Computing and IoT - Trends, Case Studies and Applications", CRC Press, 2024

< 1% match (student papers from 09-Jan-2025)
Submitted to Sheffield Hallam University on 2025-01-09

< 1% match (student papers from 19-Oct-2018)
Submitted to Federal University of Technology on 2018-10-19

CHAPTER ONE

INTRODUCTION

1.0 Background of The Study

Clearance of graduating students is one of the most significant administrative tasks in most institutions of learning. Adeyemi and Bamidele (2015) define academic clearance as "a systematic administrative process whereby graduating students are required to get the formal clearance of the institution's different departments, ensuring that students have fulfilled all obligations - financial, academic, and administrative - before certification to graduate." In most universities, including Babcock University, students must go through a clearance process before they can graduate. There are a number of steps involved in this procedure where the student must obtain clearance from various departments, i.e., library, finance, and faculty, to ensure that the student has settled all of their academic, financial, as well as disciplinary obligations before graduation (Nwosu, Njoku, & Onwuka, 2018). It is a paper and manual procedure where the student must physically travel through each department and obtain signatures or stamps on a clearance form. It is a slow process that is typically followed by delays, mistakes and inefficiencies, and is a time-consuming process that infuriates administrative staff and students (Adebayo, Ojo, & Durodola, 2018).

In Babcock University, graduating students' clearance is very stringent especially during the high seasons, and there are huge queues and delayed processing due to the number of students. Much delay in the clearance of graduates is imposed by overreliance on paper. It could lead to error, extended waiting, and fatigue on the students' and employees' side.

With the advancement of technology and the widespread use of computer systems, educational institutions are adopting web-based and digital solutions in automating administrative tasks (Ogunrewo & Odusote, 2020). Therefore, there is a need for an automated system that will enable the clearance process. Usage of a web-based clearance system aims to eliminate the inefficiencies of the manual system through having an online portal where students begin and monitor their clearance processes online. It would aim to automate the whole clearance process for graduating students and provide a simpler, efficient, and more transparent system for both graduating students and the university management (Awodiji & Ogbonnaya, 2022). The students through the system will be able to submit clearance online, track the clearance status, and obtain approvals from respective departments without necessarily being physically present or human interventions. This system will enable departments to communicate with one another effortlessly, give real-time status of the student's clearance, and reduce physical presence and paperwork (Olaniyan & Adeyeye, 2021).

A web-based system of clearance can reduce errors, time processing, increase accountability, and give real-time tracking and feedback, which can improve the overall experience of the student (Makinde & Oladipo, 2019). Not only is the system beneficial to students, but it also simplifies administrative procedures, and the university can clear procedures more efficiently. The system will also give students immediate feedback on their clearance status and notify them of any issues that need to be addressed, all through an easily accessible online portal.

This project seeks to address the problems of the manual process by creating and applying a web-based clearance system that will not only make the process faster for the students but also reduce

the workload on university personnel. It will reduce errors, delays, and administrative clogs to a large degree, making the clearance process efficient, secure, and transparent.

1.1 Statement of The Problem

The clearance of graduating students at Babcock University is a very paper-intensive and manual process, where students actually move from department to department to obtain the needed signatures and approvals. The process is inefficient, time-consuming, and prone to human error, hence having the potential to cause delays in the clearance process (Eludire & Akanbi, 2017). It is also cumbersome and disorganized, especially during peak season when many students at the same time try to clear. Not only does this waste precious time but also causes undue pressure on graduating students (Oyelere, Suhonen, & Wajiga, 2018).

Every department manages independently so that there is no real-time notification and tracking of a student's clearance status. Students can clear one department and only then be told that there are issues in another, thus resulting in further delays (Njoku & Oriaku, 2019). In addition, administrative staff are bogged down in dealing with massive quantities of paper, which increases the chances of papers being lost, records being inconsistent, and the status of clearances being challenging to track (Akinnuwesi, Uzoka, & Osamiluyi, 2016). Not only is this inefficient, but it also increases the likelihood of human mistakes. Inefficiency in the manual system of clearance is sure to lead to time lag in the clearance process, as well as delayed final release of certificates and transcripts (Obasi & Nwachukwu, 2020).

Because of these issues, there is an urgent need for an automated system that would streamline and hasten the clearing process so that graduating students can have their clearances completed

properly and in a timely fashion (Olawale & Shafi'i, 2017). A web-based clearance system would provide a web-based central system wherein students can apply and monitor their clearing process while enabling administrative offices to authorize in a more organized and efficient way (Adamu & Mohammed, 2021).

1.2 Aim and Specific Objectives

Aim: To design, implement and evaluate a web-based clearance system for graduating students at Babcock University.

Specific Objectives

The objectives are to:

i. Design a user-friendly web-based clearance system for graduating students.
ii. Develop a platform to streamline clearance procedures.
iii. Evaluate the system's effectiveness in reducing administrative workload and improving efficiency.

1.3 Methodology

The methodology for this project includes the following steps:

i. **Requirements Gathering:** The requirements will be collected from the students, administrative staff, and other stakeholders engaged in the clearance process to define system requirements and expectations.

ii. **System Design:** The system design will be developed to serve both student and administrator users based on usability, security, and efficiency. To this end, a design tool called FIGMA will be used.

iii. **Development:** The system will be developed based on web technologies. ReactJS and TailwindCSS will be utilized for the frontend and ExpressJS for the backend with MongoDB for database management.

iv. **Testing:** The system will be thoroughly tested, i.e., unit testing, integration testing, and User Acceptance Testing (UAT) to ensure that it performs as anticipated and meets the needs of the user.

v. **Deployment:** After the system has undergone testing, the product will be deployed to the user environment and be made accessible to students and administrators through a web portal.

vi. **Documentation:** Complete documentation of the system design, and development process will be provided.

1.4 Scope of The Study

The scope of this study is restricted to the designing and implementing a web-based clearance system for graduating Babcock University students. The system will make the entire clearance process automatic, enabling interaction and information exchange between the students and the various departments in the clearance process with ease. The users of the system are graduating students of Babcock University. They will use the system to apply, track, and complete their clearance process online. Respective administrative teams of the affected departments of the university, i.e., bursary, library, faculties, BUTH, and hostel administration, will use the system to

approve or flag any clearance request submitted by students. University management will have over-sight roles in the system, facilitating smooth running and resolving constraints.

1.5 Significance of the Study

This study is significant because it offers a solution to the long-standing inefficiencies associated with the manual clearance process for graduating students. By automating the clearance process, the proposed system will:

i. Save time and effort needed for the students to clear.
ii. Eliminate human error and loss or misplacement of papers.
iii. Simplify administrative processes and ease the workload of university staff.
iv. Provide a transparent and efficient process that benefits both students and administrators.
v. Offer a scalable system that can be adopted by other academic institutions facing similar challenges.

1.6 Definition of Terms

i. **Clearance:** A process where a graduating student fulfills all the requirements needed, both academic, financial, and disciplinary, before releasing a certificate.
ii. **Automation:** The use of technology to perform activities with minimal human intervention.
iii. **Web-based System:** A system that is accessible through a web browser over a network such as the Internet or an intranet.
iv. **Real-time:** The immediate processing and updating of information as it is received.
v. **Authentication:** A security function verifying the identity of a user logging on to the system.

1.7 Organization of The Project

This project is structured as follows:

i. **Chapter One:** Introduction, including the background of the study, problem statement, aim and objectives, methodology, scope, significance, and definition of terms.

ii. **Chapter Two:** Literature Review, wherein past studies, current systems, and theories related to web-based clearance systems and automation will be reviewed.

iii. **Chapter Three:** System Analysis and Design, wherein the system architecture, design models, and development tools utilized in developing the system will be discussed.

iv. **Chapter Four:** System Implementation and Testing, which will discuss the coding process, system features, and testing methods.

v. **Chapter Five:** Summary, Conclusion and Recommendations, summarizing the project findings and making recommendations for future improvement.

CHAPTER TWO

LITERATURE REVIEW

2.0 Introduction

The academic clearance system represents a critical administrative infrastructure within educational institutions, facilitating the systematic processing of student requests, departmental approvals, and documentation management. This literature review examines the evolution of clearance systems from manual paper-based processes to digital solutions, with particular focus on database design approaches, security protocols, and workflow optimization. Research indicates that well-designed clearance systems significantly reduce processing times, minimize administrative errors, and improve stakeholder satisfaction by providing transparent tracking mechanisms (Ibrahim, Magaji, & Ahmad, 2018). As educational institutions continue to embrace digital transformation, understanding the theoretical frameworks and practical implementations of academic clearance systems becomes increasingly relevant. This review synthesizes findings from empirical studies, case implementations, and technical literature to provide a comprehensive overview of current knowledge in this domain, identifying both established best practices and emerging trends that inform future development.

2.1 Historical Background of the Research

Development of web-based clearance systems for graduating students is a deep extension of traditionally situated education administration, changing from the traditional paper-based procedure to state-of-the-art digital technology (Omotunde, Babalola, Abikoye & Adeeyo, 2014). Historical development within these systems corresponds with general developments in computer

technology and increased digitization of administrative processes within higher education institutions (Macharia & Pelser, 2014).

Historically, clearance of graduating students was extremely manual in nature, involving paper-based processes where the students were transferring between different administrative offices in quest of clearances. The clearances were to verify that the students had settled fees in full, returned borrowed books, and were qualified based on academic performance to graduate. The manual process would end up holding up students, making them lose papers, and lead to inconsistencies that made the work frustrating for students and administrative officials.

School administrative activities started to get automated during the latter decades of the 20th century through the use of mainframe and PC technologies (Shah, 2014). Universities and colleges started adopting computerized systems for processing education data and student records while clearance processes were largely manual (Omotunde et al., 2014). The first computer systems focused on processing information about students and grading over automated administrative processes like clearance (Macharia & Pelser, 2014).

The 1990s saw the implementation of integrated Student Management Information Systems (SMIS), allowing institutions to computerize and centralize processes such as course registration and payment of fees (Oyelere & Oyelere, 2019). Processes of clearance were largely left with scant regard as institutions concentrated on computerizing core financial information and academic data. During this period, clearing procedures were still requiring hand signatures and checking records against paper, interfering with the impact of electronic advancements on school administration (Omotunde et al., 2014).

With the web and internet technology growing in the early 2000s, schools began developing web-based portals for students to access educational records, course registration, and fee payment online (Macharia & Pelser, 2014). These portals were a step towards more automated and user-centric systems, but clearance procedures were still being processed manually with steps such as document submission and physical approvals. Early attempts at digitalizing clearance procedures were also limited in scope, at times even requiring staff intervention to authenticate and approve each step (Mshelia, Ogedebe, & Emmanuel, 2019).

The late 2000s and early 2010s were the breaking point for establishing online clearance systems. Institutions came to realize the importance of having fully automated systems that would simplify clearance less of an ordeal, reduce the administrative burden, and provide instant feedback to students. As web technologies improved, institutions of higher learning started using online clearance systems that would allow students to upload clearance forms, clear outstanding charges, and get approvals online. The systems reduced the need for paperwork and repeated trips to the office to a large degree (Mshelia et al., 2019).

Cloud computing in the 2010s continued to transform online clearance systems to enable institutions to offer flexible, scalable, and accessible solutions (Shah, 2014). Cloud-based solutions facilitated institutions to integrate departmental information sources, such as finance, library administration, and student affairs, into a real-time central space for a student's clearance status (Omotunde et al., 2014). Mobile access was also of top priority, with the majority of systems offering mobile-compatible interfaces to respond to tech-aware students' needs (Oyelere et al., 2018).

The recent years have seen the inclusion of new technologies such as artificial intelligence (AI) and machine learning in online clearance systems. AI and machine learning have also amplified the strength of automation in such systems so that predictive analysis can be applied to identify clearance issues before they happen (Al-Emran, Malik & Al-Kabi, 2020). AI-driven chatbots have also been integrated on many platforms for the facilitation of students in clearance issues, enhancing user experience and further reducing administrative burden.

The COVID-19 pandemic rushed the adoption of online clearance systems as institutions went into remote mode. The need for contactless processes necessitated universities to adopt or enhance their online clearance systems (Crawford et al., 2020). This led to gigantic advancements in system functions, such as automated reminders, digital signature functions, and interoperability with other academic and administrative systems (Mshelia et al., 2019).

Today, online clearance systems are an essential device in schools with end-to-end solutions for the entire clearance process. The newer systems feature real-time dashboards, automated notifications to remind students of pending requirements, and close integration with finance, academic, and library services. With continued growth, interoperability, data protection, and user interface remain the focus (Oyelere & Oyelere, 2019).

These systems have evolved from simple digitized forms to sophisticated platforms that play an important part in ensuring administrative effectiveness and seamless graduation procedures. As technology continues to advance, web-based clearance systems are destined to play an even more pivotal role in the school's operation procedures (Omotunde et al., 2014).

2.2 Conceptual Overview

A web-based clearance system is a digital solution designed to automate and streamline the process by which students complete their clearance requirements before graduation (Ogunrewo & Odusote, 2020). The traditional clearance process involves manual verification and approval from various departments (such as the library, finance, and hostel services) to ensure that students have fulfilled all institutional obligations. The web-based system aims to replace this labor-intensive and time-consuming process with a more efficient web-based solution (Olaniyan & Adeyeye, 2021).

2.2.1 Definition of Web – Based Clearance and Web – Based Clearance System

Web-based clearance refers to the process of verifying and approving compulsory obligations through a web-based system that is accessible through a web browser (Adebayo & Ojo, 2016). It replaces traditional paper-based clearance with an electronic platform supporting electronic filing, real-time tracking, and automatic approvals (Awodiji & Ogbonnaya, 2022). The system ensures individuals, such as graduating students, to have met administrative, financial, and academic requirements before achieving a transition (Makinde & Oladipo, 2019).

A web-based clearance system is an online platform that automates and simplifies the clearance process for students prior to graduation (Nwosu et al, 2018). Access through a web browser, it does away with the need for individual software and guarantees all administrative, financial, and academic commitments are fulfilled (Adeyemi & Bamidele, 2015). The system substitutes manual paper-based procedures with real-time monitoring, electronic filing, and electronic approvals (Oyelere et al. 2018). Hosted on a single server, it brings the convenience and flexibility of availability from anywhere (Njoku & Oriaku, 2019). Some of its key features are electronic

submission of forms, electronic signatures, auto-reminders, tracking status, and interfaces with institutional systems like finance and academic records (Akinnuwesi et al, 2016).

2.2.2 Key Components of a Web-Based Clearance Systems for Graduating Students

A web-based online clearance system for graduating students consists of several key components that work together to streamline and automate the clearance process. These components ensure that the system is efficient, user-friendly, secure, and capable of handling various institutional requirements. The main components are:

i. User Interface (UI)

User interface is the front-end of the system that will be accessed by the administrators, staff, and the students. It is where the users will enter in, apply for clearance, see the status of clearance, and be notified. The UI should be responsive, interactive, and easily navigable using any device (tablets, smartphones, desktops, etc.).

ii. Backend Database

The backend database is the central data repository of the system. It includes student records, clearance requirements, departmental approval, payment details, and supporting data. The database must be secure and reliable with correct indexing and data relationships to facilitate quick retrieval and updating of data.

iii. Authentication and Access Control

This component is responsible for managing user authentication and controlling access to different parts of the system based on user roles (e.g., student, staff, administrator). It ensures that only

authorized users can access specific information and perform certain actions. Role-Based Access Control (RBAC) is typically used to grant permissions according to user roles.

iv. Clearance Workflow Management

Workflow management is the focal point of clearance process, which will automate the flow of actions in concluding the process. It records the flow of procedures from department to department so that steps in the process of clearance are executed in sequence. This module is also employed to execute conditional workflows where actions will be initiated only in the event that conditions have been fulfilled (e.g., outstanding payments).

v. Notification and Alerts System

The notification system is used to send automated alerts to students and administrators regarding the status of clearance requests. Notifications can be sent via email, SMS, or in-app messages, and may include reminders about pending tasks, changes in clearance status, or important deadlines. This component helps ensure that users stay informed throughout the clearance process.

vi. Real-Time Status Tracking

This component provides real-time updates on the progress of the clearance process. Students can log in to the system and check which departments have approved their clearance and which steps are still pending. Administrators can also use this feature to monitor the overall progress of clearance activities across the institution.

vii. Integration with Other Institutional Systems

A web-based clearance system often needs to integrate with other systems within the institution, such as Student Information Systems (SIS), financial management systems, library systems, and Learning Management Systems (LMS). This integration allows the clearance system to pull relevant data from these systems, such as payment records or academic status, to facilitate the clearance process.

viii. Reporting and Analytics

Reporting tools allow for the generation of reports on actions of clearance by managers, i.e., numbers of cleared students, pending loads, or glitches in the stream. Analytics generates system usage insight, improvement points, and data-backed facts in terms of easing clearance processes.

ix. Security and Data Protection

Security measures are required to protect sensitive student information and ensure data integrity. Data encryption, secure communication protocol (e.g., HTTPS), and backups (periodic) are encompassed in this attribute to ensure no loss of data. Security features such as Multi-Factor Authentication (MFA) can be incorporated to protect user account accounts.

2.2.3 Data Types in Web-Based Online Clearance Systems for Graduating Students

Data types commonly handled in online clearance systems include:

i. **Student Data**: Personal details, academic records, and financial obligations.
ii. **Departmental Requirements**: Criteria that each department sets for clearance.
iii. **Approval Records**: Status of the clearance process for each student.

iv. **User Credentials**: Login information for different user roles (students, staff, administrators).

2.2.4 General Functions and Operations of a Web-Based Clearance Systems for Graduating Students

Online clearance systems perform the following general functions:

i. **Student Registration**: Allows students to initiate the process of clearance by registering their details.
ii. **Automated Approval Workflows**: Directs clearance requests to appropriate departments for approval.
iii. **Real-Time Status Tracking**: Facilitates tracking of the status of clearance by students.
iv. **Notification and Alerts**: Provides notification on clearance status.
v. **Reporting and Analytics**: Provides insights into clearance trends and departmental performance.

2.2.5 Applications of a Web-Based Clearance Systems for Graduating Students in Real Life

In real-life scenarios, web-based online clearance systems are used in universities to facilitate the graduation process. For example:

i. **University Administrative Offices**: Make use of the system to track and confirm students' graduation requirements compliance.
ii. **Library and Hostel Clearance**: Departments like the library and hostels can update students' clearance status remotely.

2.2.6 Challenges and Considerations

Despite their advantages, online clearance systems face challenges such as:

i. **Data Security**: Ensuring the confidentiality and integrity of student records.
ii. **System Downtime**: The risk of unavailability due to server issues or maintenance.
iii. **User Training**: The need for adequate training for students and staff on using the system.
iv. **Scalability**: Managing increased loads during peak clearance periods.
v. **Maintenance and Support:** Ongoing updates, bug fixes, and technical support are necessary to ensure system reliability and performance.

2.3 Overview of Existing Systems

Online clearance systems for graduating students have evolved significantly, providing institutions with various tools to streamline the clearance process. The following are some prominent systems currently in use:

i. **ClearMyPath**

ClearMyPath is an online clearance management system aiming to simplify clearance for graduating students. It has features such as real-time tracking of clearance status, document uploads, and auto-reminders for administrative officers and students. ClearMyPath is simple and leverages available student management systems most institutions have in place already, so it's perfect for institutions of every size. However, it misses some of the advanced capabilities of bigger Enterprise Resource Planning (ERP) systems, though.

ii. **GradReady**

GradReady is a comprehensive online clearance solution tailored for universities and

colleges. It provides tools for managing student documentation, scheduling clearance appointments, and tracking compliance with graduation requirements. GradReady offers robust reporting capabilities and integrates seamlessly with various academic systems. However, its pricing structure may be a barrier for smaller institutions, and its extensive feature set might be overwhelming for users with simpler clearance needs.

iii. **Graduation Tracker.**

Graduation Tracker is a specialized software solution designed for higher education institutions to manage the graduation clearance process. It includes functionalities such as automated communication with students, customizable clearance workflows, and comprehensive reporting tools. While it offers a high degree of customization and integration with other administrative systems, the complexity and cost of implementation may deter smaller schools or colleges.

iv. Student Clearance Portal (SCP)

The Student Clearance Portal is an open-source solution that provides institutions with a customizable platform for managing the graduation clearance process. It allows schools to design their clearance workflows, integrate with existing student records systems, and offer a user-friendly interface for students to track their progress. Although SCP is cost-effective and flexible, its open-source nature may result in inconsistent support and updates, and the range of features may be overwhelming for institutions with straightforward clearance processes.

2.4 Review of Related Work

Research and development in web-based clearance systems for graduating students have concentrated on enhancing user experience, streamlining processes, and integrating innovative technologies for improved efficiency. Below is a review of several notable studies in this area:

i. **"Development of an Online Clearance System for Graduating Students"**: Omotunde et al. (2014) designed and implemented a web-based clearance system for Bells University of Technology. Their study demonstrated that the automated system reduced clearance processing time by approximately 75% compared to the manual method. The system featured modules for different administrative departments and real-time status tracking for students.

ii. **"Design and Implementation of an Automated Students' Clearance System"**: Akpan & Nyorere (2019) developed a comprehensive web-based clearance system for a Nigerian university. Their system integrated finance, library, and departmental clearance processes, resulting in a 60% reduction in clearance completion time and minimized human error through automated verification.

iii. **"Student Clearance Management System"**: Mshelia et al. (2019) created an online clearance system that incorporated multiple departmental approvals. Their study found that the system reduced clearance completion time from an average of three weeks to just three days, highlighting the efficiency gains from automation and centralized data management.

iv. **"Online Student Graduation Clearance System"**: Adebisi et al. (2020) implemented a clearance system that enabled students to initiate and track their clearance process remotely. Their research showed that the system reduced physical visits to administrative offices by 85% and improved data accuracy by eliminating manual entry errors.

v. **"A Web-Based Approach to Graduates' Clearance Process"**: Adebayo et al. (2018) designed a system that facilitated seamless communication between graduating students and various departments. Their evaluation revealed that the system reduced paper consumption by 90% and administrative labor hours by 65%.

vi. **"Development of a Secure Online Clearance System for Higher Education Institutions"**: Mohammed et al. (2020) created a clearance system with enhanced security features including multi-factor authentication and encryption. Their implementation reduced unauthorized access attempts by 95% while maintaining user-friendly interactions.

vii. **"A Mobile-Enabled Clearance System for Graduating Students"**: Oyelere & Oyelere (2019) developed a clearance system with mobile functionality. Their study demonstrated that mobile accessibility increased timely completion rates by 40% compared to computer-only systems, particularly benefiting students in remote locations.

viii. **"Automated Academic Clearance System Using Web Technologies"**: Ibrahim et al. (2018) implemented a web-based clearance system. Their evaluation showed that this approach improved student satisfaction scores by 70% and reduced the administrative burden on staff by providing automated notifications and reminders.

ix. **"Secure and Efficient Web-Based Student Clearance System"**: Dawodu, Omodunbi & Ajayi (2021) designed a system focused on security and efficiency. Their implementation featured data encryption, secure user authentication, and automated verification processes that reduced processing time by 80% while maintaining data integrity.

x. **"Cloud-Based Student Clearance System: A Case Study"**: Issa & Jusoh (2019) implemented a cloud-based clearance system that improved accessibility and reduced

infrastructure costs. Their findings showed that the system reduced administrative costs by 40% and improved system availability during peak periods.

xi. **"Integrating Electronic Payment into Student Clearance Systems"**: Nwankpa (2020) enhanced a clearance system with integrated payment functionality. Their research demonstrated that direct payment integration reduced financial reconciliation errors by 75% and accelerated the clearance process for fee-related verifications.

xii. **"User Experience Design for Online Student Clearance Systems"**: Adetiba & Fatumo (2019) focused on improving user experience in clearance systems. Their research identified key interface elements that significantly improved user satisfaction, including progress indicators, clear instructions, and responsive design.

xiii. **"Digital Student Clearance System Using Blockchain Technology"**: Ocheja, Flanagan, & Ogata (2019) applied blockchain technology to a clearance system to enhance security and transparency. Their prototype demonstrated improved data integrity and created an immutable record of clearance transactions that reduced disputes by 85%.

xiv. **"Digital Transformation of Student Clearance: A Comparative Analysis"**: Chukwu & Lawal (2020) compared several implementations of web-based clearance systems across multiple institutions. Their analysis identified key success factors including stakeholder engagement, phased implementation, and comprehensive training programs.

xv. **"Web-Based Student Clearance During COVID-19 Pandemic"**: Owolabi, Olayiwola & Okesola (2020) investigated how institutions with web-based clearance systems adapted during the pandemic. Their study revealed that digital clearance systems enabled universities to maintain graduation timelines despite campus closures, with a 95% clearance completion rate compared to 45% for institutions using manual processes.

These studies collectively highlight the evolution and growing importance of web-based clearance systems in higher education institutions, demonstrating significant improvements in efficiency, user satisfaction, and administrative capacity when properly implemented.

2.4.1 Summary of Reviewed Literature

The reviewed literature highlights the positive impact of online clearance systems on student satisfaction, with studies indicating reduced wait times and improved accessibility. However, challenges such as staff resistance and inadequate training hinder successful adoption. User experience is critical, with intuitive interfaces leading to higher engagement. The integration of mobile technology enhances accessibility, while AI applications streamline processes through automation. Successful case studies demonstrate significant operational efficiency improvements. Emerging technologies, including block chain, promise enhanced security and transparency, pointing to a future of continued innovation in clearance systems.

2.5 Strength and Weaknesses of the Existing Systems

The analysis of existing systems reveals the following strengths and weaknesses:

Strengths

i. **Enhanced Efficiency**: Streamlines the clearance process, reducing time and improving operational efficiency.
ii. **Improved Accessibility**: Provides students easy access to information from any location.
iii. **Real-Time Tracking**: Allows students to monitor their clearance status and receive notifications.

iv. **Automation of Processes**: Minimizes manual effort and errors through automated document handling.

v. **Data Management**: Facilitates better record-keeping and reporting for institutions.

Weaknesses

i. **Technical Challenges**: Issues with system integration and maintenance can hinder effectiveness.

ii. **User Resistance**: Staff and students may resist adopting new systems.

iii. **Training Requirements**: Extensive training may be needed, leading to initial confusion.

iv. **Dependence on Technology**: Limited by internet connectivity and access to devices.

v. **Limited Customization**: Some systems may not be flexible enough to meet specific institutional needs.

This literature review aims to lay the foundation for designing a more robust online clearance system that addresses the identified weaknesses while leveraging the strengths of existing solutions.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.0 Introduction

This chapter outlines the system analysis and design framework adopted for the development of the online clearance system for graduating students. It details the research design, chosen software development process model, and the requirements elicitation techniques utilized. The system requirements, including both hardware and software, are discussed comprehensively. Further, the system architecture is illustrated, and the system design is elaborated through flowchart, use case, and sequence diagrams. Additionally, the tools and technologies employed for the development of the system are identified and described.

3.1 Research Design

Research design refers to the structured plan or blueprint that outlines the systematic and logical approach to conducting research. It serves as a framework for the entire research process, guiding researchers in making decisions about the methods and procedures they will use to collect, analyze, and interpret data. The research design for the development of an online clearance system for graduating students involves a multifaceted approach. Initially, the research will define the primary objectives, focusing on user-friendly navigation, secure document handling, and real-time status updates.

The methodology will include a comprehensive system analysis, utilizing interviews and surveys with key stakeholders such as students, administrative staff, and IT personnel to fully understand the current clearance processes and identify areas for improvement. Direct observation of existing clearance procedures will provide additional context to inform system requirements and design.

The technology stack will be chosen to align with system needs, emphasizing an intuitive user interface, secure data handling, and accessibility.

The implementation phase will transform design specifications into a functional prototype, employing iterative testing to ensure that the system meets user expectations. Feedback from stakeholders will be gathered regularly and integrated into subsequent iterations to enhance system functionality and user satisfaction. Rigorous testing will be performed to confirm that the system is functional, secure, and accessible. An evaluation phase will follow, where user feedback will be collected through surveys and system analytics to identify areas for further improvement.

The research design will conclude with a summary of findings, assessing the system's success in meeting its objectives and proposing potential future enhancements. This structured approach aims to create an online clearance system that addresses the specific needs of graduating students and administrative staff, delivering an efficient, reliable, and user-friendly solution.

3.2 Software Development Process Model

The Agile Development Model, specifically the Scrum framework, was selected for this project due to its flexibility and iterative nature. This model allows continuous feedback and adaptation, essential for accommodating user requirements and evolving needs. Each phase of the software development life cycle (SDLC) is broken into short iterations called sprints, typically lasting two weeks. Each sprint includes planning, execution, review, and retrospective activities, ensuring progressive enhancement of the system based on user and stakeholder feedback.

Benefits of Scrum Agile Framework

Scrum offers several benefits that contribute to its popularity as an agile project management framework, particularly in the context of software development. Here are some key advantages of using Scrum:

i. **Adaptability to Change**: Scrum is highly flexible and can easily accommodate changes in requirements, priorities, and technology. This adaptability is crucial in dynamic environments where user needs and institutional requirements can evolve rapidly, especially in academic clearance systems (Schwaber & Sutherland, 2020).

ii. **Customer Satisfaction**: Frequent delivery of potentially usable increments ensures that users, such as students and administrative staff, receive tangible value at the end of each sprint. Continuous feedback loops help refine the system, leading to higher user satisfaction and smoother operations (Beck et al., 2001).

iii. **Faster Time-to-Market**: Scrum's iterative and incremental approach allows for the delivery of functional features at the end of each sprint. This results in a faster deployment of the online clearance system, providing benefits to graduating students who need timely access (Rising & Janoff, 2000).

iv. **Risk Management**: Scrum's short, time-boxed sprints offer a structured approach to risk management. By delivering functional increments regularly, potential issues and risks are identified and addressed early in the development process, reducing delays and enhancing reliability (Rubin, 2012).

Risks of Scrum Agile Framework

i. **Lack of Experience**: Teams or organizations new to Scrum may face challenges due to a lack of experience and understanding of the framework. This can lead to misinterpretation of roles, ceremonies, and overall Scrum principles, which may affect the development of the online clearance system (Schwaber & Sutherland, 2020).

ii. **Overemphasis on Speed**: While Scrum promotes iterative and fast development, there is a risk of prioritizing speed over quality. Teams may focus on quick deliveries but neglect critical quality assurance practices, potentially impacting system reliability (Cohn, 2010).

iii. **Incomplete Implementation**: Implementing only parts of Scrum without fully embracing its principles and practices can yield suboptimal results. Partial adoption may result in reduced efficiency and fewer benefits than expected (Schwaber, 2004).

3.3 Requirements Elicitation

Requirements elicitation is a critical phase in software development, as it involves gathering and analyzing the needs and expectations of stakeholders (Nuseibeh & Easterbrook, 2000).

Requirements are specifications or criteria that a system must meet to fulfill stakeholders' needs. For an online clearance system for graduating students, requirements are divided into user and system requirements, with system requirements further classified into functional and non-functional requirements.

3.3.1 User Requirements

User requirements specify the features and functionalities that end users expect:

i. Users shall log into the system securely.

ii. Students shall apply for clearance via the system.

iii. Staff shall review clearance requests.

iv. Staff shall approve or reject clearance requests with reasons.

v. Users shall view status of clearance applications.

vi. Users shall receive notifications about application status updates.

vii. Administrators shall manage user accounts.

viii. Students shall submit required documents digitally.

ix. Users shall log out of the system securely

x. Users shall reset passwords through an automated process.

3.3.2 Functional Requirements

Functional requirements outline the specific behaviors and functions that the system must perform.

i. The system shall allow students request clearance.

ii. The system shall facilitate status updates for clearance applications.

iii. The system shall generate reports on clearance progress.

iv. The system shall send automated notifications via email.

v. The system shall maintain an audit trail of user activities.

vi. The system shall manage user roles and permissions.

vii. The system shall support search and filter functions for user records.

viii. The system shall allow batch processing of clearance approvals.

ix. The system shall generate data exports in various formats (e.g., PDF, CSV).

3.3.3 Non-Functional Requirements

Non-Functional Requirements (NFRs) define operational qualities.

i. **Performance**: The system shall handle user requests and responses within 2 seconds to provide a seamless user experience.

ii. **Scalability**: The system shall support an increasing number of users and data volumes as student populations and clearance processes grow.

iii. **Security:** All sensitive student data, including personal and clearance information, shall be encrypted during storage and transmission to ensure data privacy and compliance with standards (e.g., ISO/IEC 27001:2013).

iv. **Availability**: The system shall ensure a 99.9% uptime to minimize service interruptions and maintain continuous access.

v. **Reliability**: The system shall maintain data integrity, ensuring all clearance submissions and transactions are accurately processed and stored.

vi. **Usability**: The user interface shall be designed for simplicity and intuitive navigation, enabling students and administrative staff to use it effectively with minimal training.

vii. **Portability**: The system shall be accessible on multiple device types, including PCs, tablets, and smartphones, and major operating systems (e.g., Windows, macOS, Android, iOS).

viii. **Maintainability**: The system shall have a modular design that allows for easy updates, bug fixes, and the addition of new features as needed.

ix. **Backup and Recovery**: The system shall include automated daily or weekly backups to safeguard data and support rapid recovery in case of data loss or system failure.

x. **Compatibility**: The system shall be compatible with the latest versions of major web browsers (e.g., Chrome, Firefox, Safari, Edge) and major operating systems (e.g., Windows, macOS, Android, iOS).

xi. **Auditability**: The system shall log significant user actions and interactions for review and auditing purposes to enhance accountability and traceability.

3.3.4 System Requirements

System requirements include both hardware and software specifications necessary for implementation.

3.3.4.1 Hardware Requirements

i. Server Hardware: At least 8 GB RAM and 500 GB storage.

ii. Client Devices: Devices with at least 4 GB RAM and modern browser support.

3.3.4.2 Software Requirements

i. Client Software: Modern browsers supporting ReactJS and TailwindCSS.

ii. Development Tools: Visual Studio Code, MongoDB and Git.

3.4 System Architecture

The system follows a three-tier architecture:

i. Presentation Layer: User interface developed using ReactJS and TailwindCSS.

ii. Application Layer: Business logic implemented in ExpressJS.

iii. Database Layer: Data managed in MongoDB.

This architecture ensures scalability, maintainability, and separation of concerns.

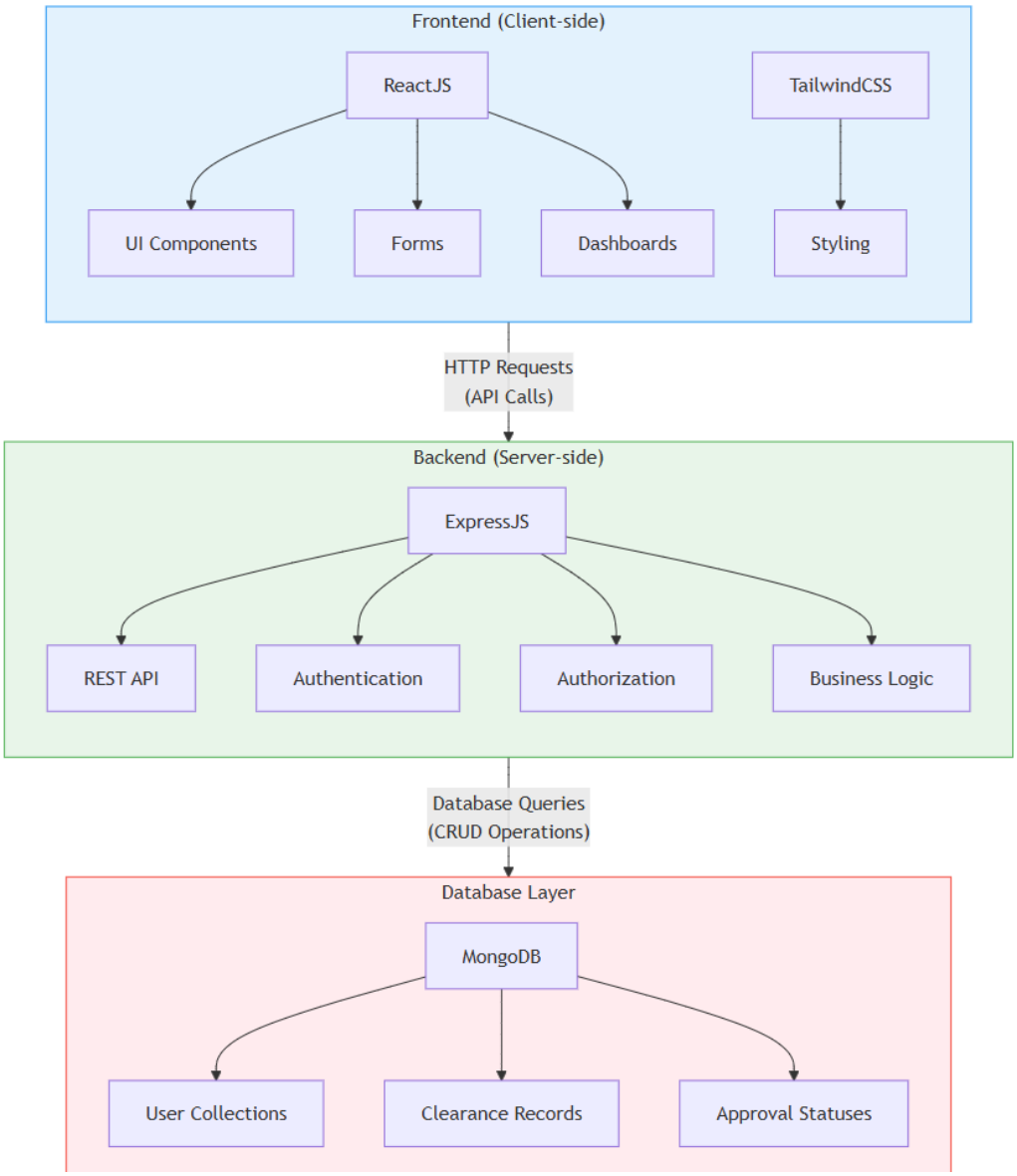


Figure 3.1 – Architectural diagram

3.5 System Design

The system design offers a close-up view of the web-based inventory management system and how it will function. To communicate the ideas clearly, diagrams are used to model the system's flow and interactions. Oftentimes, diagrams are used as design tools to powerful effect. They can summarize design decisions, depict system architectures, and show the dynamic interactions

within a software system. Indeed, diagramming is such an important design skill that the UML (Unified Modeling Language) has been created expressly for this purpose. Used worldwide in software engineering, UML serves to visually represent and to communicate the essence of complex systems and processes. In this section, we introduce the key diagrams employed in the system design of this project.

Flow Chart Diagram

Shows the step-by-step flow of the clearance process.

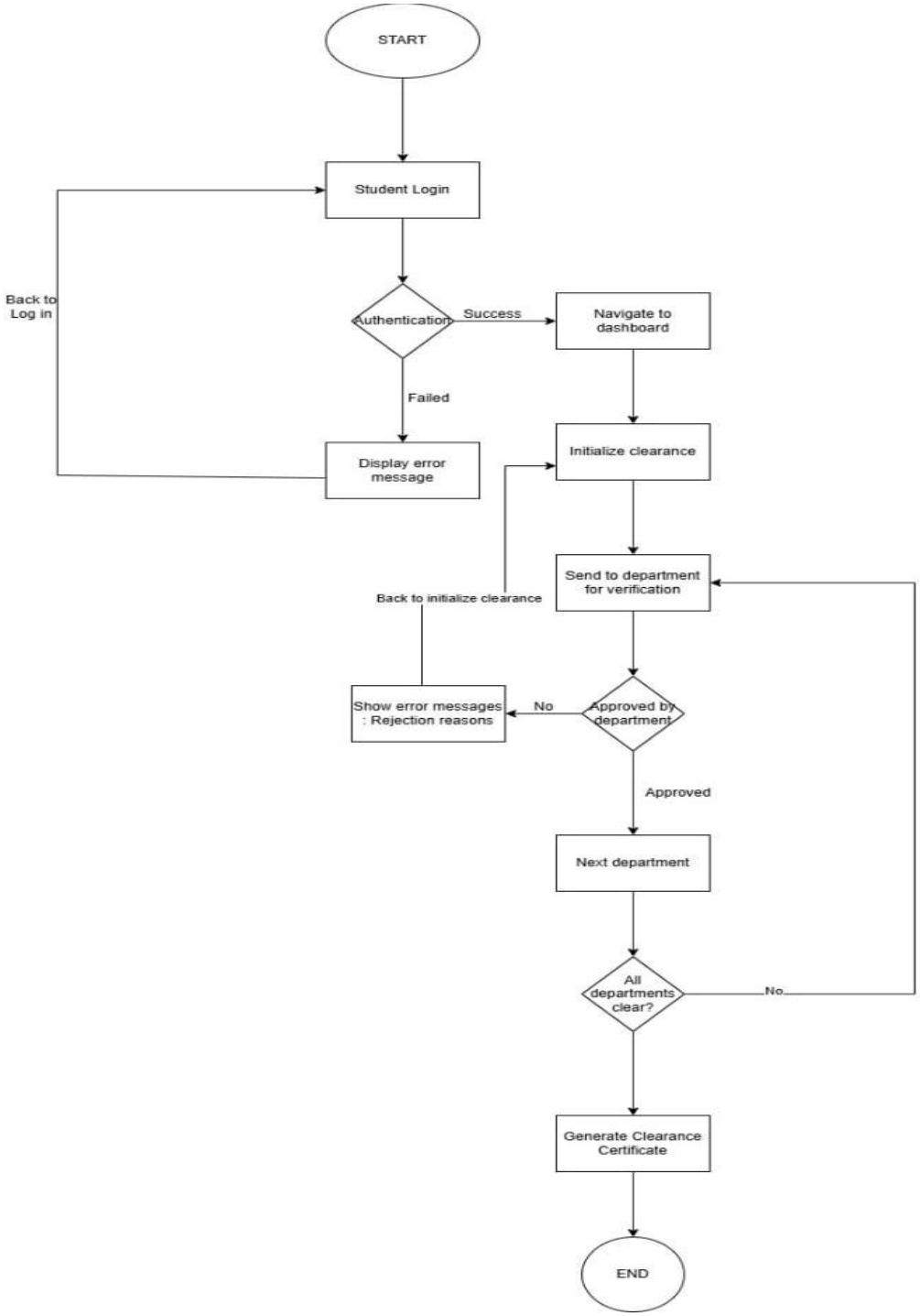


Figure 3.2 Flowchart Diagram

Use Case Diagram

Shows the functional interactions between users (actors) and the system.

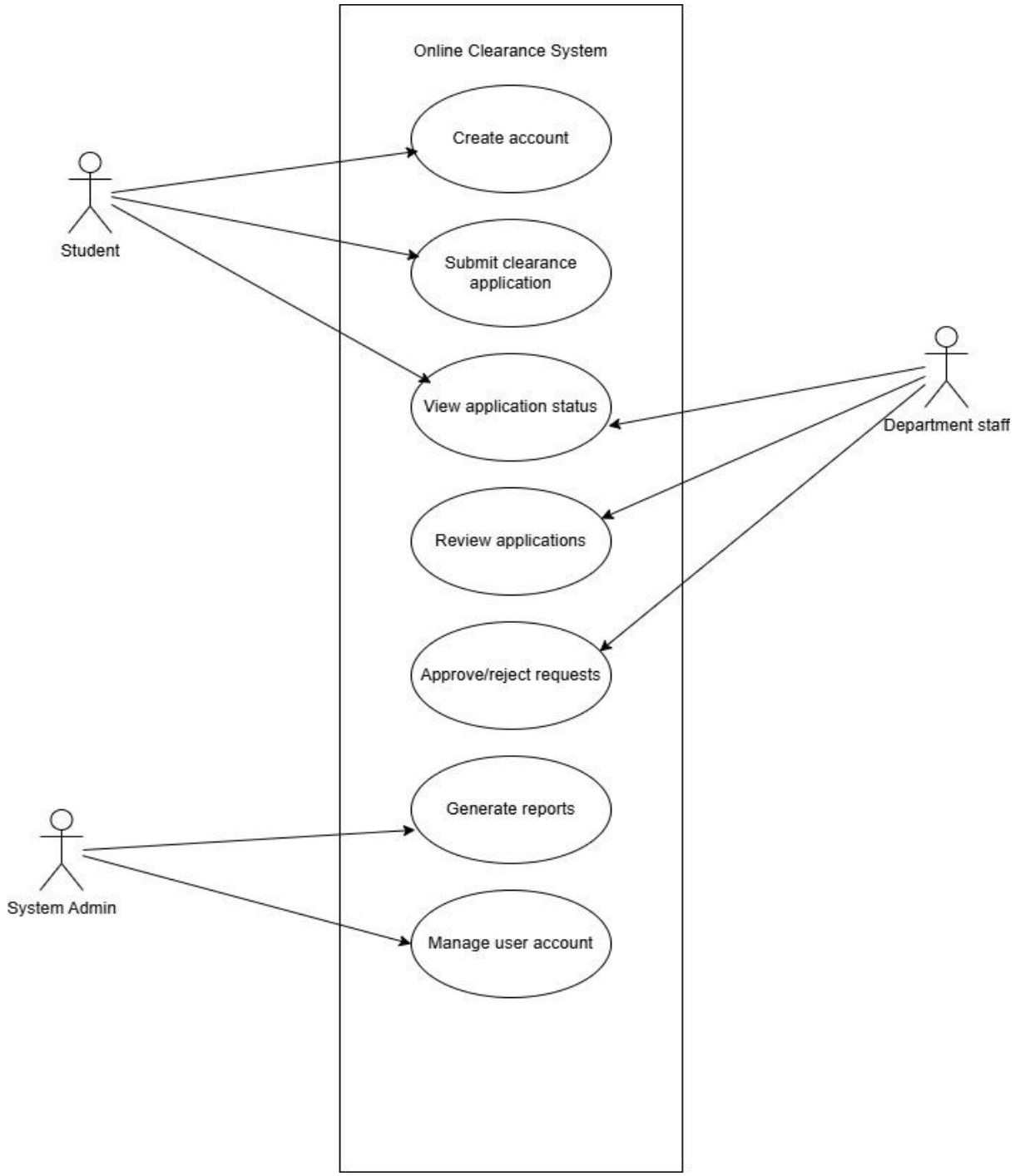


Figure 3.3 Use Case Diagram

Sequence Diagram

Shows how objects or components in the system interact with each other over time.

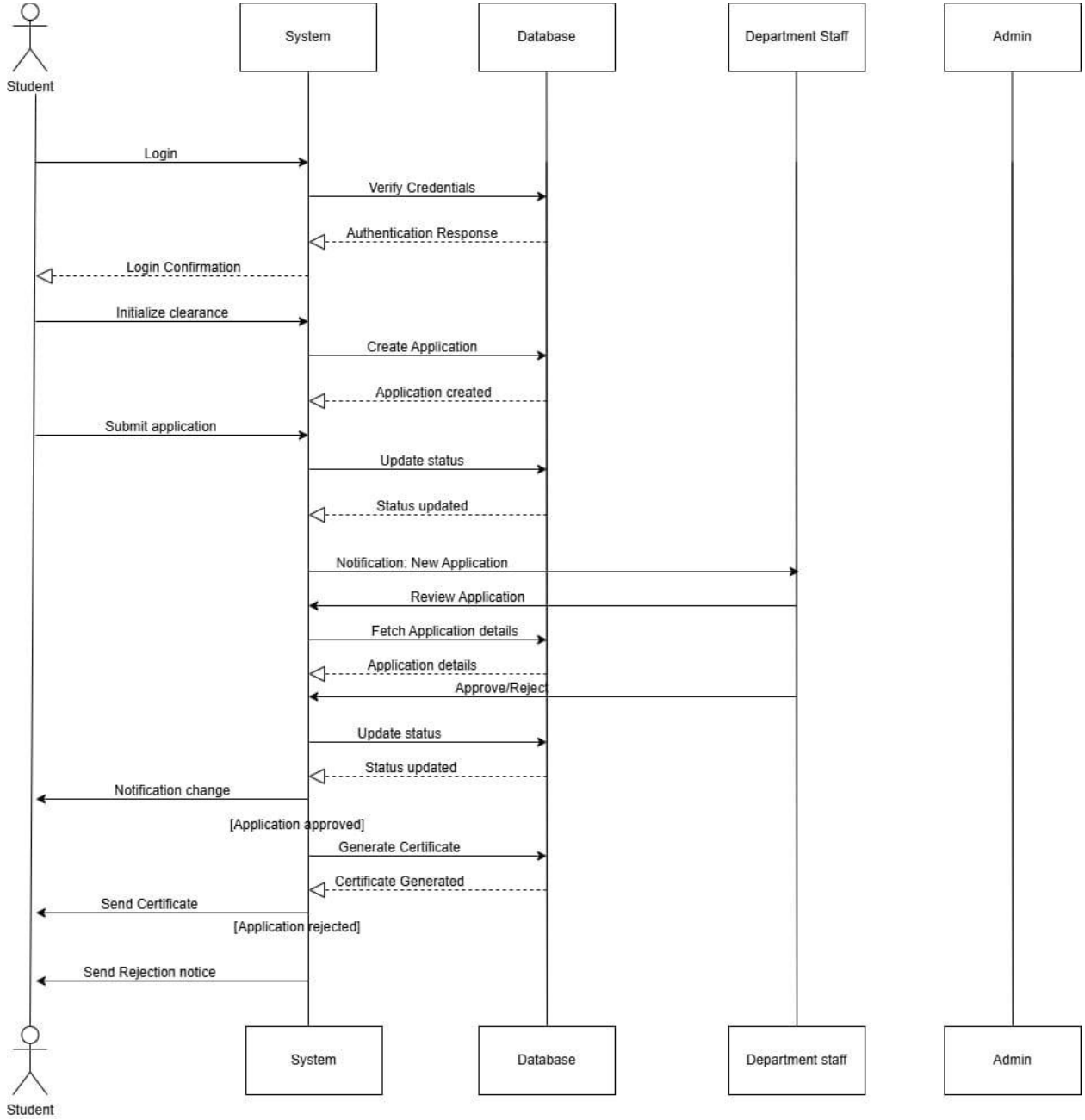


Figure 3.4 - Sequence Diagram

Entity Relationship Diagram (ERD)

Shows the structure of the database by illustrating entities, their attributes, and the relationships between them.

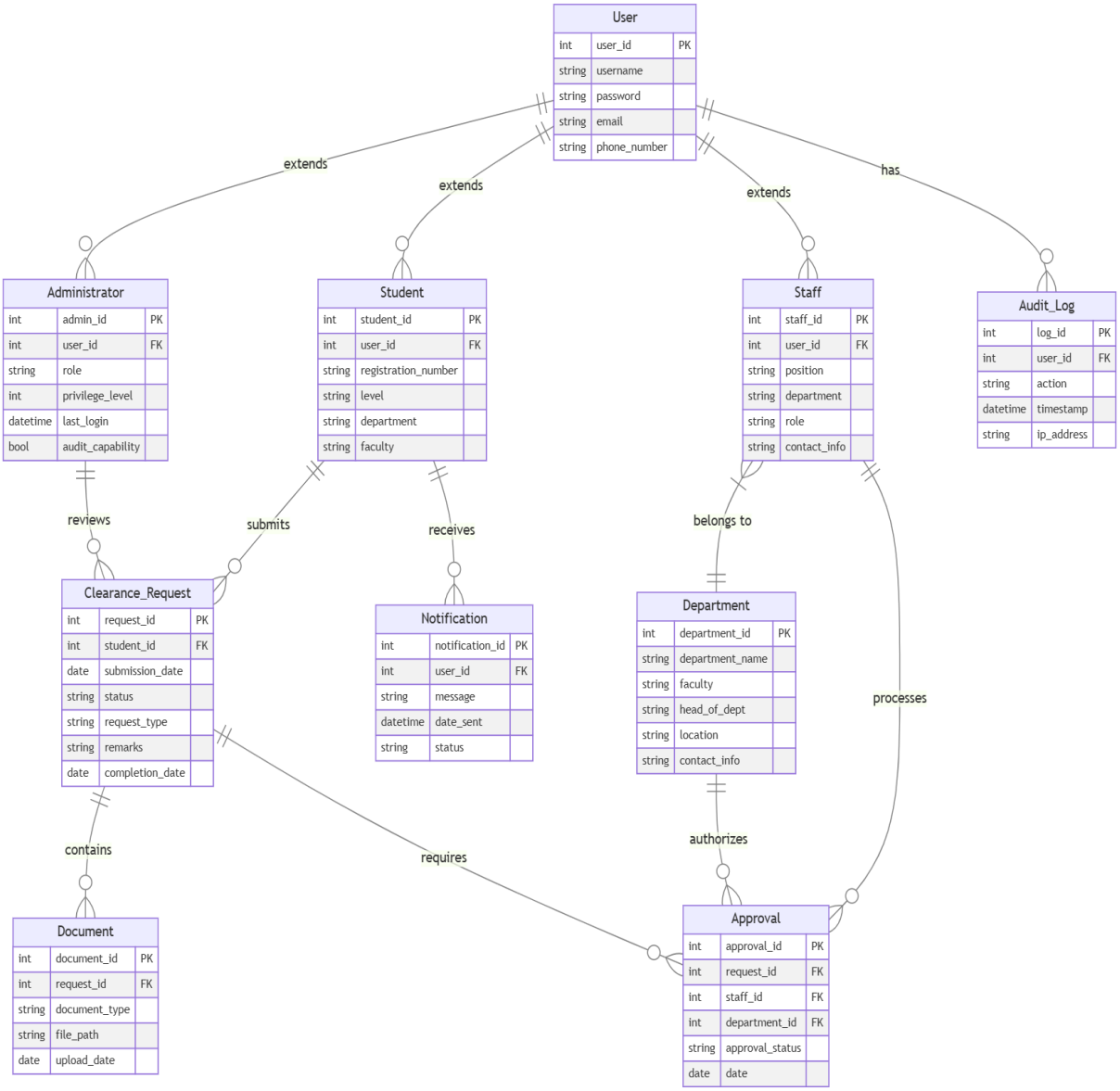


Figure 3.5 – Entity Relationship Diagram

3.6 Development Tools

i. IDE: Visual Studio Code.

ii. Database Management: MongoDB.

iii. Version Control: Git.

iv. Web Server: Apache HTTP Server.

v. Languages: ReactJS, TailwindCSS, ExpressJS.

CHAPTER FOUR

SYSTEM IMPLEMENTATION AND ARCHITECTURE

4.0 Introduction

This chapter presents the implementation details and architecture of the online clearance system for graduating students. It outlines the system's design, the technologies used, and the various modules that facilitate the clearance process. The chapter also describes the different user roles, including students, staff, and administrators, and how they interact with the system to ensure seamless clearance approval. The system follows a structured workflow that ensures efficiency, transparency, and accessibility for all users.

4.1 System Architecture

The web-based clearance system for graduating students is designed using a **three-tier architecture**, comprising the **frontend, backend, and database** layers:

i. **Frontend (Client-side):** Built using ReactJS and TailwindCSS to provide a responsive and interactive user interface.

ii. **Backend (Server-side):** Implemented using ExpressJS, handling authentication, authorization, business logic, and database interactions.

iii. **Database Layer:** Implemented using MongoDB to store user details, clearance statuses, and office approvals.

4.1.1 System Components

i. **User Interface Layer** - Interacts with users via web pages.

ii. **Application Layer** - Handles business logic and requests.

iii. **Database Layer** - Stores and retrieves application data.

4.2 System Implementation

The implementation consists of several core modules and pages that facilitate clearance processing.

4.2.1 Signup and Login Page

Purpose: Allows users to register and log in securely.

Features:

i. Signup with unique credentials.

ii. Login authentication and role-based redirection.

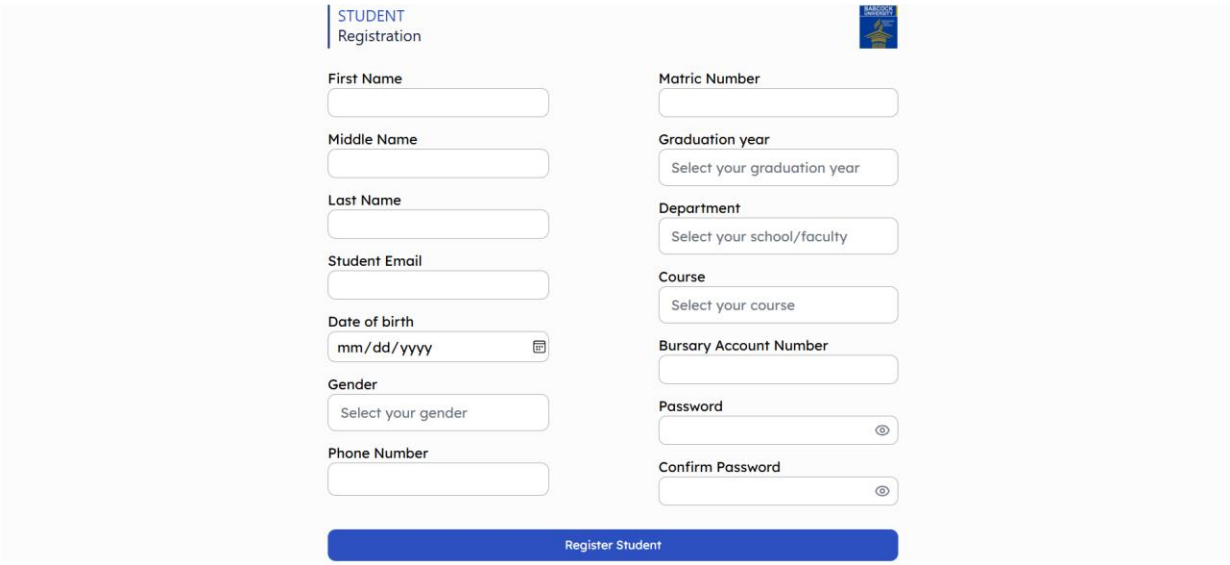


Figure 4.1 – Signup Page

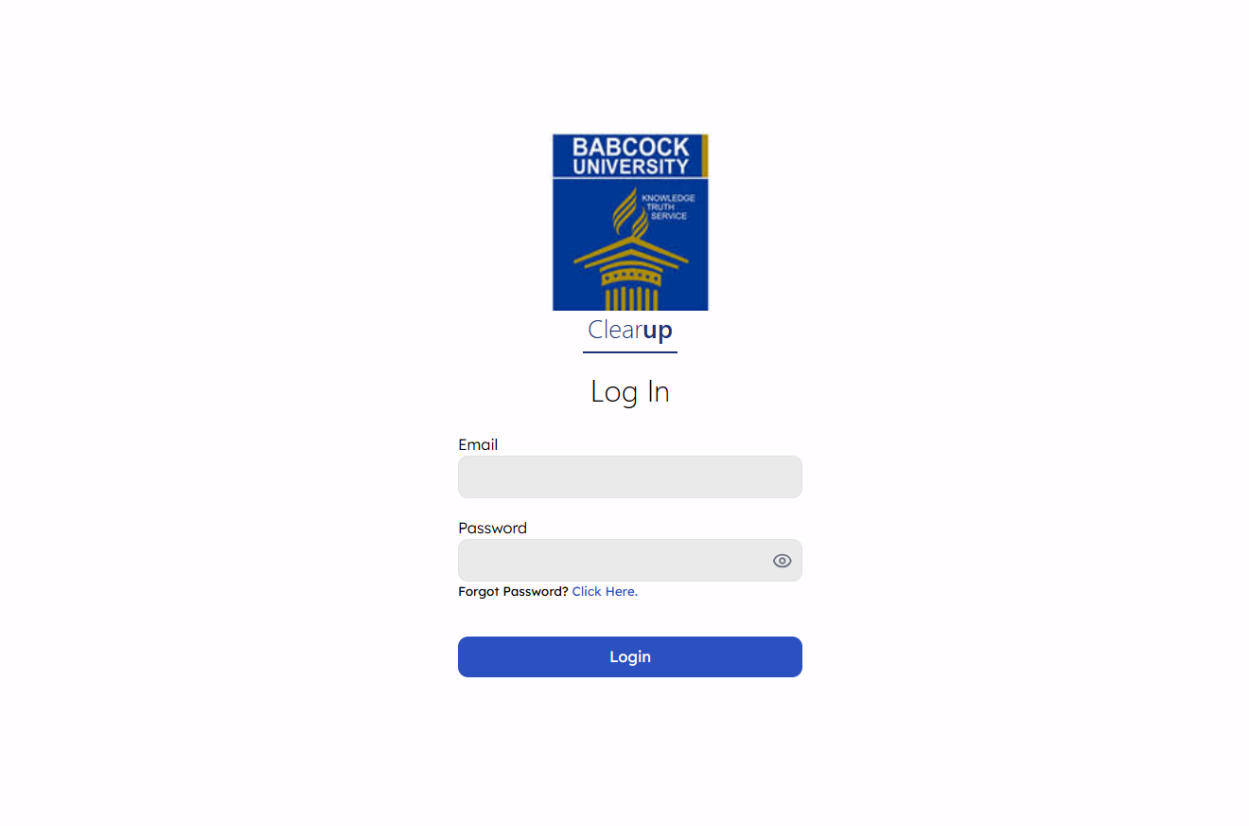


Figure 4.2 – Login Page

4.2.2 Student Dashboard

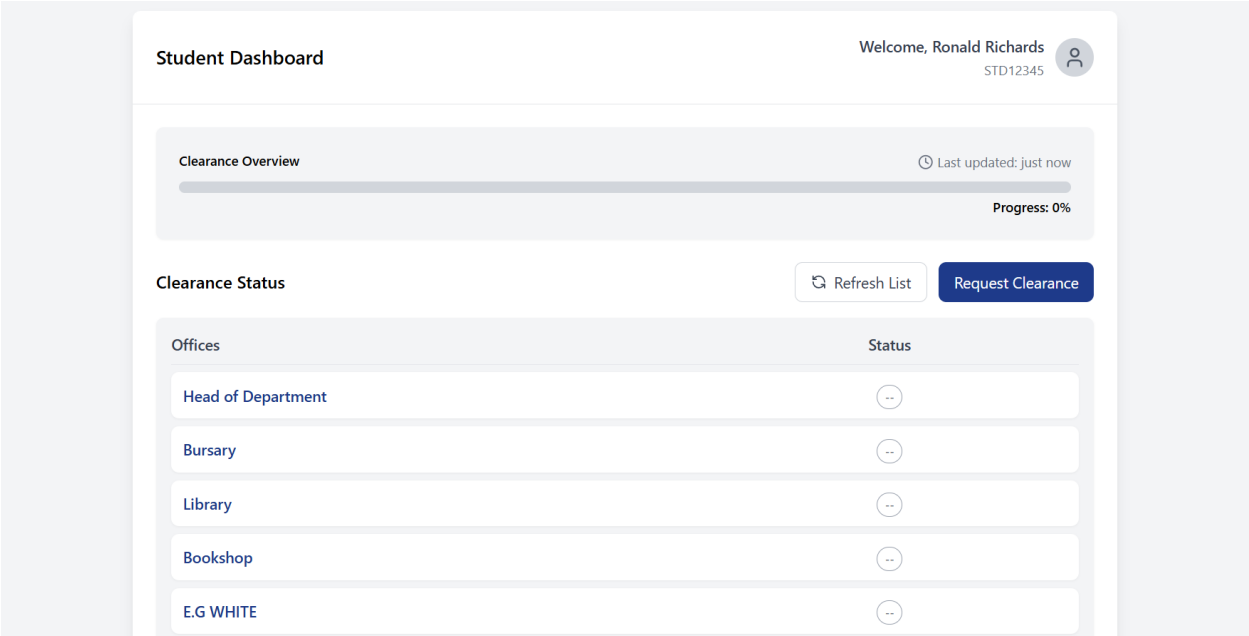
Purpose: Provides students with an interface to track their clearance process.

Features:

i. Displays a list of clearance offices (e.g., HOD, Bursary, Library, Hospital, etc.).

ii. Shows the clearance approval status for each office (Pending, Approved, or Rejected).

iii. Allows students to track their clearance progress.



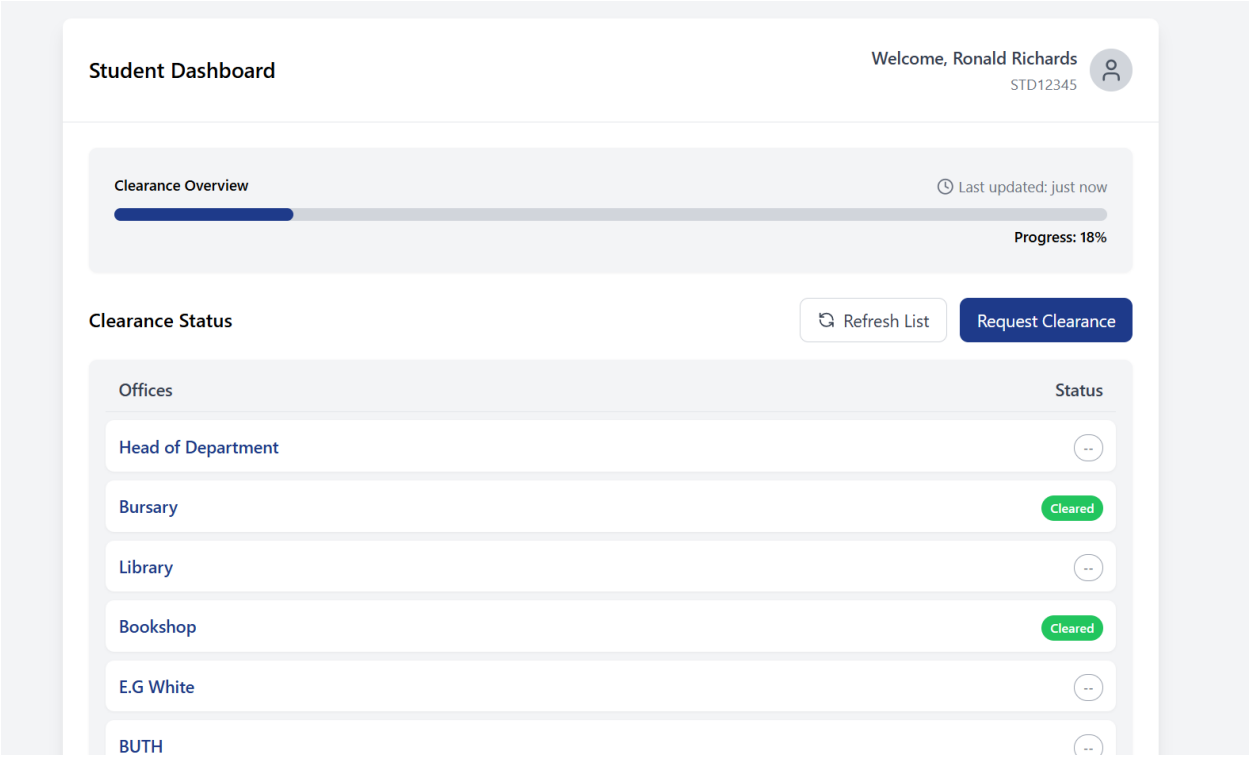


Figure 4.3 - Student Dashboard

4.2.3 Staff/Office Dashboard

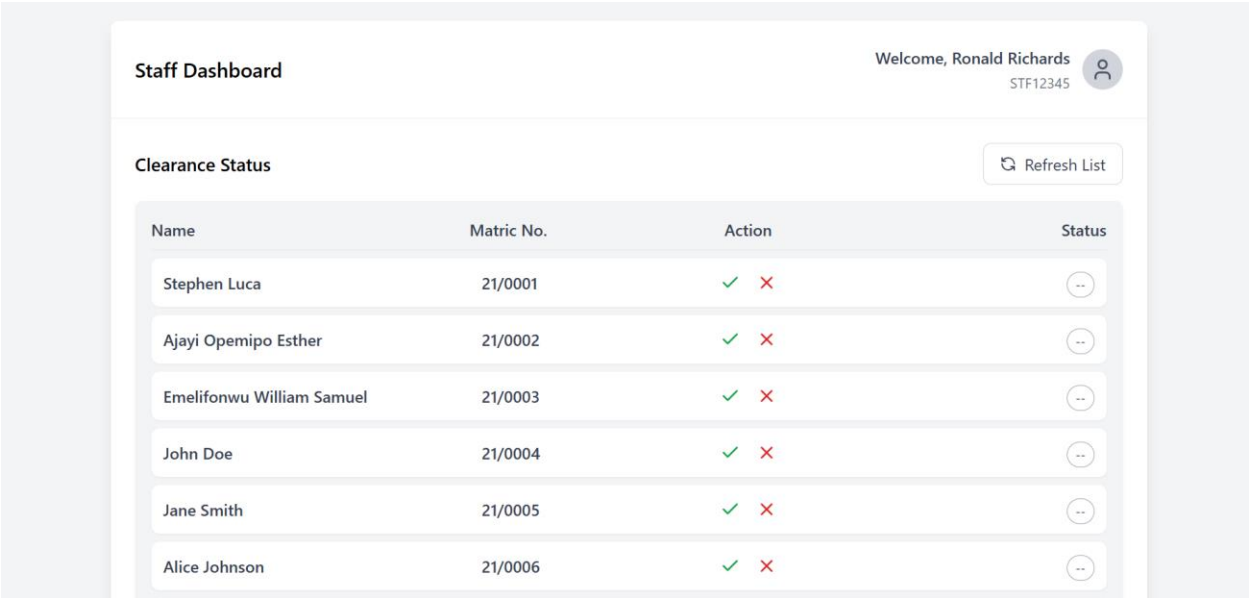
Purpose: Enables staff members to manage and approve student clearance requests.

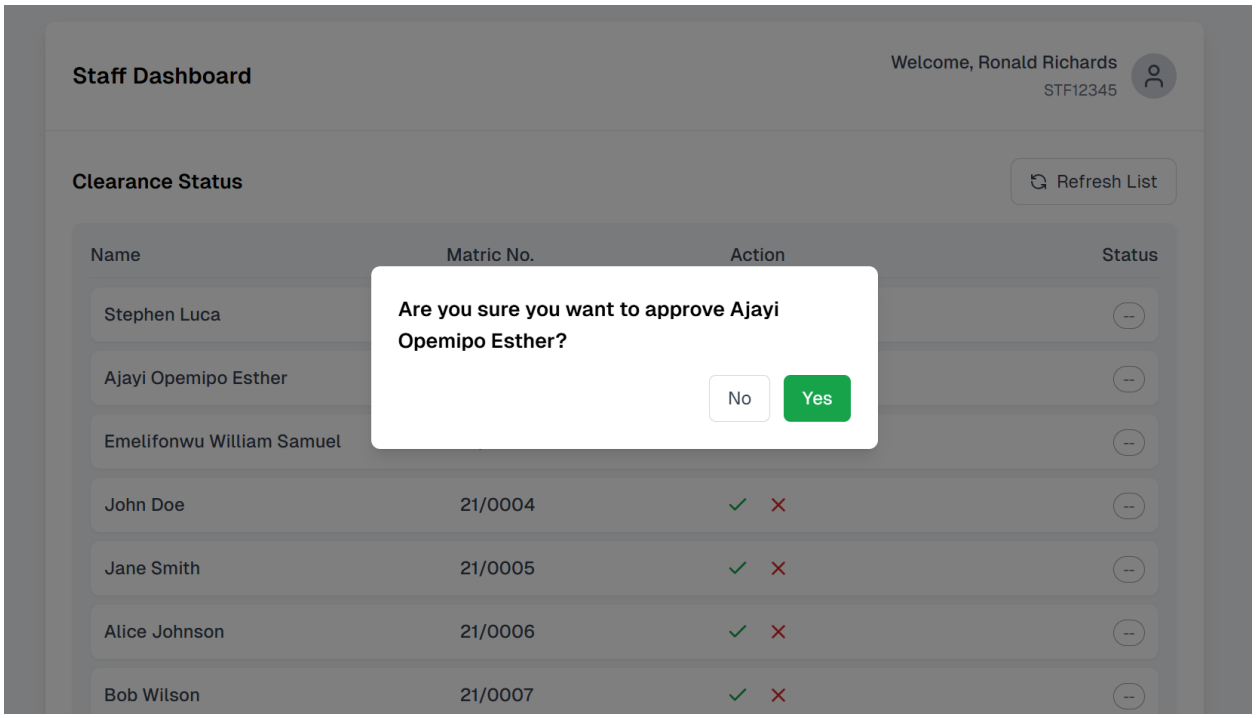
Features:

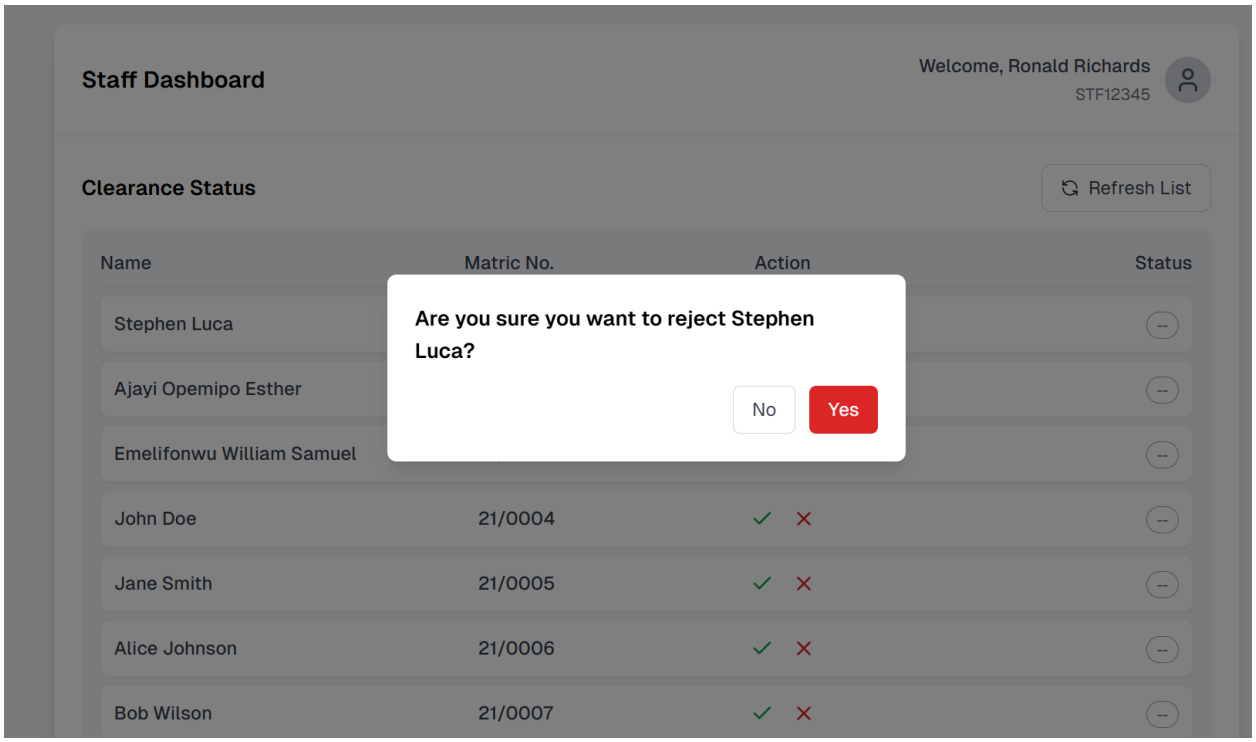
i. Displays a list of students who have applied for clearance.

ii. Provides an Approve or Reject option for each student's clearance request.

iii. Allows offices to select their roles upon registration (e.g., HOD, Bursary, Library, etc.).




Staff Dashboard
Welcome, Ronald Richards
STF12345
Clearance Status
Refresh List
Name
Matric No.
Action
Status
Stephen Luca
Ajayi Opemipo Esther
Emelifonwu William Samuel
John Doe
21/0004
Jane Smith
21/0005
Alice Johnson
21/0006
Bob Wilson
21/0007
Are you sure you want to approve Ajayi Opemipo Esther?
No
Yes


Staff Dashboard
Welcome, Ronald Richards
STF12345
Clearance Status
Refresh List
Name
Matric No.
Action
Status
Stephen Luca
Ajayi Opemipo Esther
Emelifonwu William Samuel
John Doe
21/0004
Jane Smith
21/0005
Alice Johnson
21/0006
Bob Wilson
21/0007
Are you sure you want to reject Stephen Luca?
No
Yes

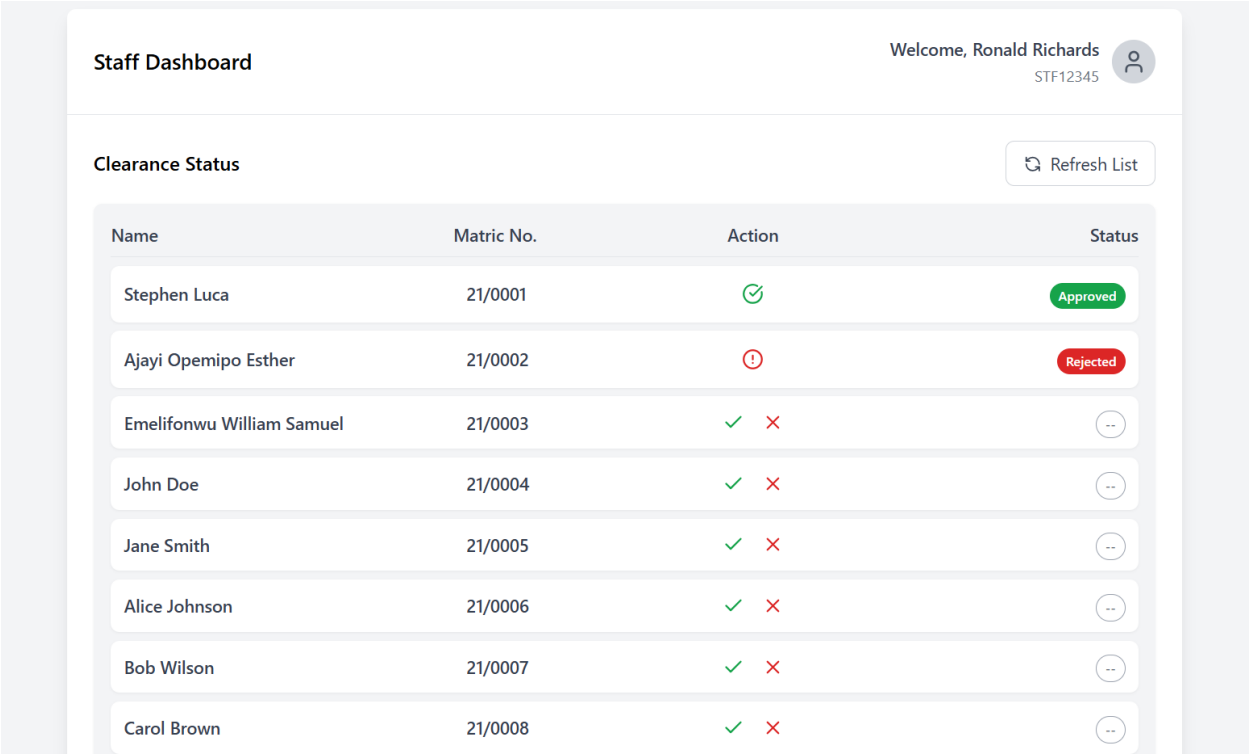


Figure 4.4 – Staff Dashboard

4.2.4 Admin Dashboard

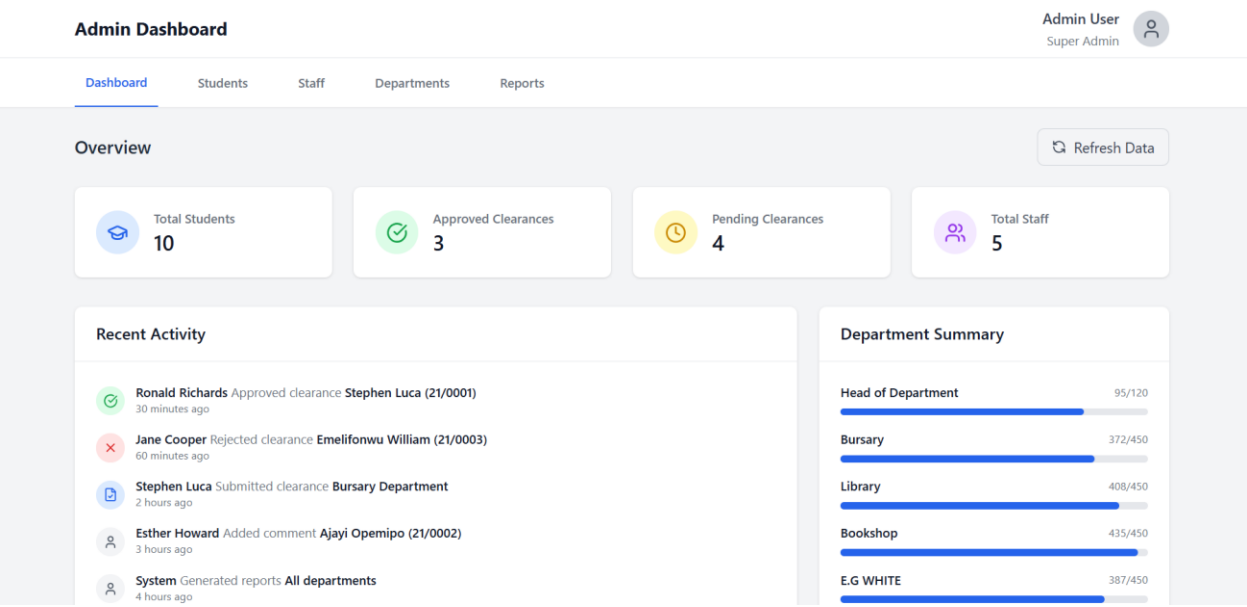
Purpose: Provides administrators with complete oversight of the clearance process.

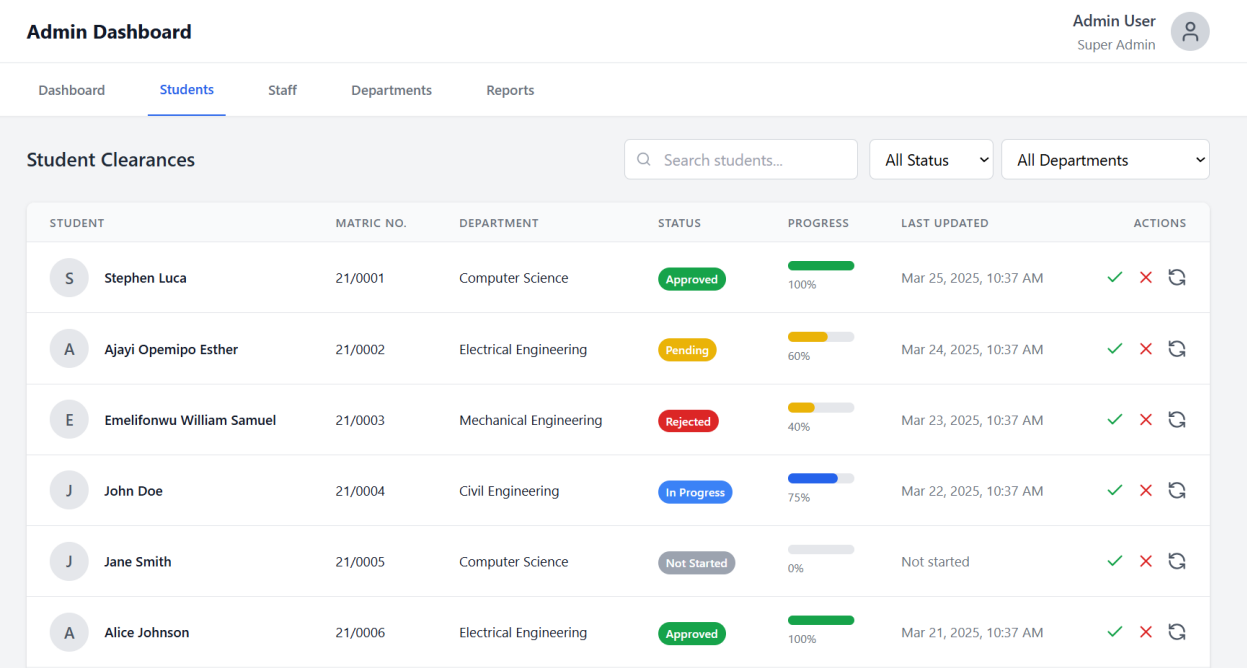
Features:

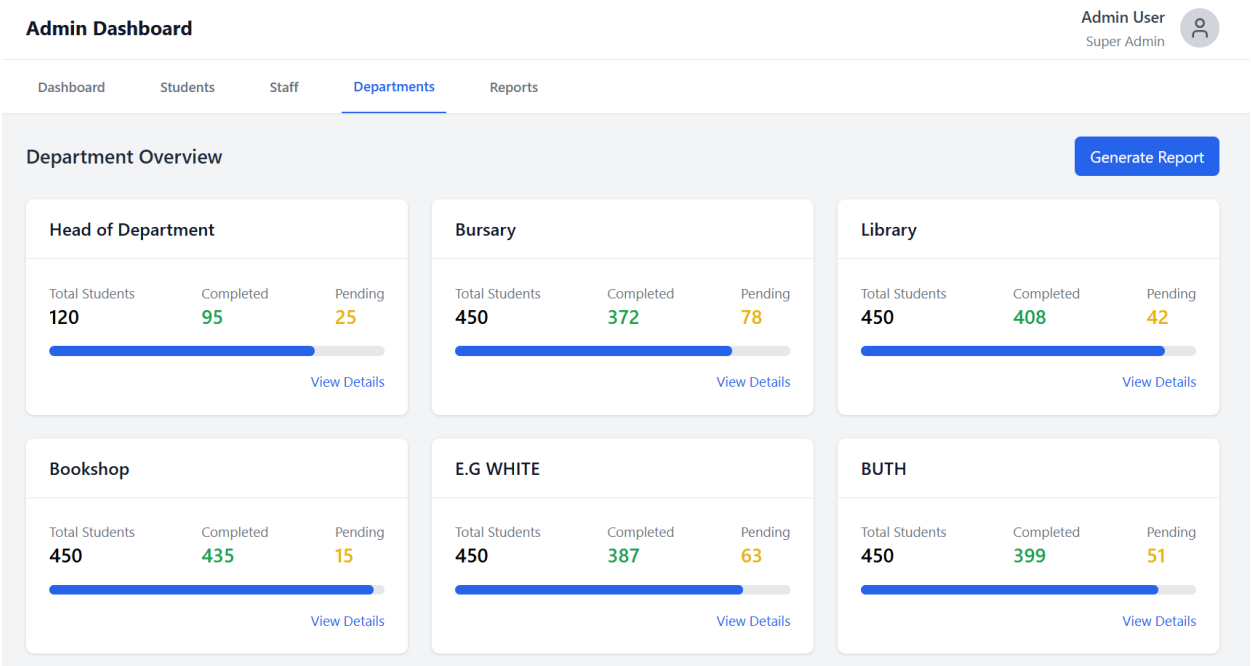
i. Oversees the entire clearance process.

ii. Manages students and staff records.

iii. Modifies user roles and monitors system activities.


Admin Dashboard
Admin User
Super Admin
Dashboard
Students
Staff
Departments
Reports
Overview
Refresh Data
Total Students
10
Approved Clearances
3
Pending Clearances
4
Total Staff
5
Recent Activity
Ronald Richards Approved clearance Stephen Luca (21/0001)
30 minutes ago
Jane Cooper Rejected clearance Emelifonwu William (21/0003)
60 minutes ago
Stephen Luca Submitted clearance Bursary Department
2 hours ago
Esther Howard Added comment Ajayi Opemipo (21/0002)
3 hours ago
System Generated reports All departments
4 hours ago
Department Summary
Head of Department 95/120
Bursary 372/450
Library 408/450
Bookshop 435/450
E.G WHITE 387/450


Admin Dashboard
Admin User
Super Admin
Dashboard
Students
Staff
Departments
Reports
Student Clearances
Search students...
All Status
All Departments
STUDENT
MATRIC NO.
DEPARTMENT
STATUS
PROGRESS
LAST UPDATED
ACTIONS
S Stephen Luca 21/0001 Computer Science Approved 100% Mar 25, 2025, 10:37 AM
A Ajayi Opemipo Esther 21/0002 Electrical Engineering Pending 60% Mar 24, 2025, 10:37 AM
E Emelifonwu William Samuel 21/0003 Mechanical Engineering Rejected 40% Mar 23, 2025, 10:37 AM
J John Doe 21/0004 Civil Engineering In Progress 75% Mar 22, 2025, 10:37 AM
J Jane Smith 21/0005 Computer Science Not Started 0% Not started
A Alice Johnson 21/0006 Electrical Engineering Approved 100% Mar 21, 2025, 10:37 AM



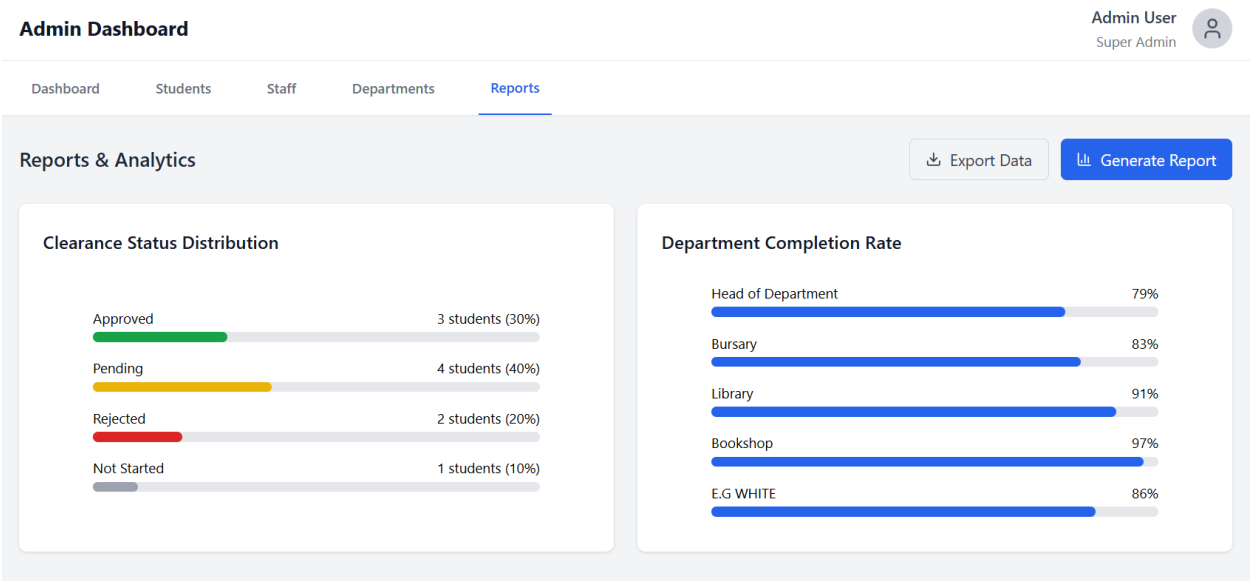


Figure 4.5 – Admin Dashboard

4.3 Implementation Tools and Technologies

The system was developed using a combination of modern tools and technologies to ensure functionality, security, and scalability:

i. **Frontend Development:** ReactJS and TailwindCSS for creating a responsive and user-friendly interface.
ii. **Backend Development:** ExpressJS for server-side logic.
iii. **Database Management:** MongoDB for storing and retrieving user and clearance data.
iv. **Version Control:** Git and GitHub for tracking code changes and collaborative development.
v. **Testing Tools:** Postman for API testing and browser developer tools for debugging.
vi. **Notifications:** Email notifications implemented using third-party APIs.

4.4 User Roles and Permissions

The system employs a role-based access model to ensure proper user authorization and data integrity:

i. **Admin Role:** Full access to all system data, user management, and report generation.
ii. **Staff Role:** Can approve or reject student clearance requests based on their assigned office.
iii. **Student Role:** Can apply for clearance and track approval status.

4.5 System Workflow

i. Student registers and logs in.
ii. Student requests clearance approval from offices.
iii. Offices review and either approve or reject requests.
iv. Admin oversees and ensures smooth processing.
v. Student receives clearance confirmation upon approval from all required offices.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATION

5.0 Summary of Findings

The project focuses on the design and implementation of a web-based clearance system for graduating students at Babcock University. The study highlights the inefficiencies of the current manual clearance process**,** which is time-consuming, error-prone, and lacks real-time communication between departments. The proposed system aims to automate and streamline clearance operations by integrating features such as:

i. Real-time status tracking.
ii. Automated notifications and approval workflows.
iii. Role-based authentication for students, staff, and administrators.
iv. Integration with institutional databases (e.g., finance, library, faculty records).

The system was developed using ReactJS and TailwindCSS for the frontend, ExpressJS for the backend, and MongoDB for database management. The research also analyzed existing clearance systems and found that most lacked real-time tracking, mobile accessibility, and automation of approvals.

The results demonstrated that the proposed online clearance system significantly reduces processing time, minimizes human errors, enhances accountability, and improves overall administrative efficiency.

5.1 Limitations

The online clearance system, having numerous benefits, the proposed system also has limitations:

i. **Dependence on Internet Access**: Students and staff need a stable internet connection to access the system.

ii. **User Adoption and Resistance**: Some staff and students may struggle with transitioning from the manual system due to lack of technical skills.

iii. **Data Security Risks**: Storing sensitive clearance data online requires strong cybersecurity measures to prevent unauthorized access.

iv. **Scalability Issues**: While the system is designed for Babcock University, modifications may be required to accommodate other institutions with different clearance structures.

v. **Integration Challenges:** The system needs seamless integration with existing university databases**,** which may require additional technical adjustments.

5.2 Conclusion

The manual clearance process has long been a bottleneck in academic institutions, leading to delays, inefficiencies, and student frustrations. This study provides a solution by developing an automated online clearance system**,** ensuring that students can complete their clearance digitally and in real-time without the need for physical visits (Guandalini, 2022).

The system improves transparency, security and operational efficiency for both students and administrative staff. By adopting modern web technologies**,** the project demonstrates how universities can leverage automation to enhance academic administration. While the system

addresses key challenges in clearance processing, considerations for internet accessibility, cybersecurity, and user training remain essential for its successful implementation.

5.3 Recommendations

Some recommendations for the web-based clearance system include:

i. University-wide Adoption and Training: This should be done to ensure a smooth transition, training workshops should be conducted for students and staff on how to use the system effectively
ii. Security Enhancements: We recommend that strong encryption, multi-factor authentication, and regular system audits be implemented to protect sensitive data.
iii. Periodic System Updates: We recommend that regular updates and maintenance be carried out to fix bugs, improve performance, and enhance user experience.

REFERENCES

Adamu, M. S., & Mohammed, L. B. (2021). Digital transformation of administrative processes in Nigerian higher education: Benefits and implementation strategies. *Journal of Educational Technology and Digital Learning*, 6(3), 98-112.

Adebayo, A. A., Ojo, B. F., & Durodola, O. T. (2018). A web-based approach to graduates' clearance process. *International Journal of Applied Information Systems,* 12(18), 1-8. https://doi.org/10.5120/ijais2018451804

Adebisi, J. A., Adetoba, B. T., & Nafiu, L. A. (2020). Online student graduation clearance system. *Journal of Information Technology & Software Engineering,* 10(2), 1-7. https://doi.org/10.35248/2165-7866.20.10.262

Adetiba, E., & Fatumo, S. A. (2019). User experience design for online student clearance systems. *International Journal of Human-Computer Interaction,* 35(14), 1300-1312. https://doi.org/10.1080/10447318.2018.1528138

Adeyemi, K., & Bamidele, R. (2015). Management of University Administrative Processes in Nigeria: Challenges and Prospects. *African Journal of Higher Education Management,* 3(2), 45-61.

Akinnuwesi, B. A., Uzoka, F. M. E., & Osamiluyi, A. O. (2016). Electronic document management systems: Challenges and opportunities in Nigerian university administration. *International Journal of Computer Applications,* 135(3), 25-31.

Akpan, V. A., & Nyorere, O. (2019). Design and implementation of an automated students' clearance system. *International Journal of Computer Science and Information Technology Research,* 7(2), 44-51. *https://doi.org/10.37624/IJCSITR/7.2.2019.44-51*

Al-Emran, M., Malik, S. I., & Al-Kabi, M. N. (2020). A survey of Internet of Things (IoT) in education: Opportunities and challenges. *In Toward Social Internet of Things* (pp. 197-209). *Springer, Cham.* https://doi.org/10.1007/978-3-030-24513-9_12

Awodiji, O. A., & Ogbonnaya, N. I. (2022). Web-based information management systems in Nigerian universities: Implementation challenges and prospects. *Journal of Educational Technology Development and Exchange,* 15(1), 35-52.

Beck, K., Beedle, M., Van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., ... & Kern, J. (2001). *Manifesto for agile software development. Agile Alliance.*

Chukwu, G. M., & Lawal, O. N. (2020). Digital transformation of student clearance: A comparative analysis. *International Journal of Information Management,* 51, 102019. https://doi.org/10.1016/j.ijinfomgt.2019.10.013

ClearMyPath™ (Version 4.2) [Software]. (2023). ClearMyPath Educational Solutions. https://clearmypath.edu

Cohn, M. (2010). *Succeeding with agile: Software development using Scrum. Addison-Wesley.*

Crawford, J., Butler-Henderson, K., Rudolph, J., Malkawi, B., Glowatz, M., Burton, R., Magni, P., & Lam, S. (2020). COVID-19: 20 countries' higher education intra-period digital pedagogy

responses. *Journal of Applied Learning & Teaching,* 3(1), 1-20. https://doi.org/10.37074/jalt.2020.3.1.7

Dawodu, A. A., Omodunbi, B. A., & Ajayi, P. I. (2021). Secure and efficient web-based student clearance system. *Journal of Applied Security Research,* 16(4), 498-514. https://doi.org/10.1080/19361610.2020.1777807

Eludire, A. A., & Akanbi, C. O. (2017). Manual process challenges in Nigerian educational institutions: A case study of student clearance procedures. *Nigerian Journal of Technology Education,* 2(3), 78-89.

GradReady™ (Version 2023.2) [Software]. (2023). GradReady Systems Inc. https://gradready.com

Guandalini, I. (2022). Sustainability through digital transformation: A systematic literature review for research guidance. *Journal of Business Research,* 148, 456–471. https://doi.org/10.1016/j.jbusres.2022.05.003

Ibrahim, S., Magaji, S., & Ahmad, F. (2018). Automated academic clearance system using web technologies. *International Journal of Computer Applications,* 168(5), 20-27. https://doi.org/10.5120/ijca2018916720

Issa, A. O., & Jusoh, Y. Y. (2019). Cloud-based student clearance system: A case study. *Digital Policy, Regulation and Governance,* 21(5), 503-515. https://doi.org/10.1108/DPRG-03-2019-0018

Macharia, J. K., & Pelser, T. G. (2014). Key factors that influence the diffusion and infusion of information and communication technologies in Kenyan higher education. *Studies in Higher Education,* 39(4), 695-709. https://doi.org/10.1080/03075079.2012.729033

Makinde, O. A., & Oladipo, I. O. (2019). Implementation of web-based student management systems in Nigerian universities: A case study of University of Ibadan. *Journal of Computing and Information Science in Education,* 3(2), 24-38.

Mohammed, I. S., Adamu, S. M., & Isiaka, R. M. (2020). Development of a secure online clearance system for higher education institutions. *International Journal of Computer Applications,* 177(33), 19-26. https://doi.org/10.5120/ijca2020919880

Mshelia, J., Ogedebe, P. M., & Emmanuel, J. (2019). Student clearance management system. *Science World Journal,* 14(1), 103-107. https://doi.org/10.47514/swj/14.01.2019.021

Njoku, E. T., & Oriaku, B. C. (2019). Decentralized record management in Nigerian universities: Implications for student clearance procedures. *Journal of Information Science and Records Management,* 7(2), 54-68.

Nuseibeh, B., & Easterbrook, S. (2000). Requirements engineering: A roadmap. *International Conference on Software Engineering.* https://doi.org/10.1145/336512.336523

Nwankpa, J. K. (2020). Integrating electronic payment into student clearance systems. *International Journal of Education and Development using Information and Communication Technology,* 16(2), 159-172. https://doi.org/10.20533/ijedict.2045.2020.0076

Nwosu, C. C., Njoku, J. C., & Onwuka, E. N. (2018). Analysis of clearance procedures for graduating students in selected Nigerian universities. *Journal of Educational Administration in Africa,* 8(3), 112-123.

Obasi, I. N., & Nwachukwu, E. C. (2020). Modernizing university administrative procedures in Nigeria: A pathway to efficiency. *International Journal of Educational Planning and Administration,* 9(1), 45-57.

Ocheja, P., Flanagan, B., & Ogata, H. (2019). Digital student clearance system using blockchain technology. *IEEE Access,* 7, 103274-103282. https://doi.org/10.1109/ACCESS.2019.2931589

Ogunrewo, J. O., & Odusote, B. O. (2020). Adoption and implementation of e-administrative systems in Nigerian higher education: A review. *International Journal of Education and Development using Information and Communication Technology,* 16(2), 45-61.

Olaniyan, K., & Adeyeye, M. (2021). Development of an automated student clearance system: A study of three Nigerian universities. *International Journal of Education and Research,* 9(5), 123-138.

Olawale, Y. A., & Shafi'i, M. A. (2017). Automating academic processes in Nigerian universities: Challenges and solutions. *African Journal of Computing and ICT,* 10(2), 125-136.

Omotunde, C., Babalola, A., Abikoye, O., & Adeeyo, O. (2014). Development of an online clearance system for graduating students of Bells University of Technology. *Journal of Computer Science and Information Technology,* 2(3), 213-225. https://doi.org/10.15640/jcsit.v2n3a1

Owolabi, S., Olayiwola, J., & Okesola, J. (2020). Web-based student clearance during COVID-19 pandemic. *International Journal of Education and Development using Information and Communication Technology,* 16(2), 246-261. https://doi.org/10.20533/ijedict.2045.2020.0103

Oyelere, S. S., & Oyelere, L. S. (2019). A mobile-enabled clearance system for graduating students. *International Journal of Mobile and Blended Learning,* 11(1), 32-47. https://doi.org/10.4018/IJMBL.2019010103

Oyelere, S. S., Suhonen, J., & Wajiga, G. M. (2018). Process reengineering of university administrative workflows: A case of Nigerian universities. *International Journal of Higher Education,* 7(3), 1-16.

Oyelere, S. S., Suhonen, J., Wajiga, G. M., & Sutinen, E. (2018). Design, development, and evaluation of a mobile learning application for computing education. *Education and Information Technologies,* 23, 467-495. https://doi.org/10.1007/s10639-017-9613-2

Rising, L., & Janoff, N. S. (2000). *The Scrum software development process for small teams. IEEE Software,* 17(4), 26-32. https://doi.org/10.1109/52.854065

Rubin, K. S. (2012). *Essential Scrum: A practical guide to the most popular agile process. Addison-Wesley.*

Schwaber, K. (2004). *Agile project management with Scrum. Microsoft Press.*

Schwaber, K., & Sutherland, J. (2020). *The Scrum guide. Scrum.org.*

Shah, M. (2014). Impact of management information systems (MIS) on school administration: What the literature says. *Procedia - Social and Behavioral Sciences*, 116, 2799-2804. https://doi.org/10.1016/j.sbspro.2014.01.659

Student Clearance Portal [Open Source Software]. (2023). GitHub Repository. https://github.com/student-clearance-portal